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REVIEW ARTICLE

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Heterogeneous mechanical metamaterials: design paradigms, additive manufacturing, and emergent multifunctionality

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ABSTRACT

Heterogeneous metamaterials constitute a new frontier in architected materials, combining multiphase compositions, hierarchical architectures, and coupled multi physics interactions to achieve unprecedented mechanical and functional performance. By harnessing synergistic interactions among diverse material phases, unit cell topologies, and physical mechanisms, these metamaterials enable programmable, adaptive, and multifunctional responses far beyond those of homogeneous systems. This review provides a systematic synthesis of six major design strategies, multiphase design, hybrid architectures, interpenetrating composites, multi material fusion, multifeature integration, and data driven approaches, emphasising their structure–mechanism–property relationships and nonlinear coupling effects. We critically analyse how structural heterogeneity governs deformation evolution, energy dissipation, and cross field functionality, and highlight their broad potential in impact protection, flexible electronics, acoustic control, adaptive actuation, and thermal management. Finally, we identify key challenges in correlating multiscale structures with emergent functions, improving manufacturing scalability, and developing unified modelling frameworks. By bridging fundamental design principles with engineering applications, this review outlines the roadmap towards intelligent heterogeneous metamaterials, offering fresh perspectives for next generation multifunctional systems.

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1. Introduction

With the rapid progress of advanced manufacturing technologies and the introduction of bioinspired design principles [1–5], mechanical metamaterials, artificially architected structures whose properties arise primarily from geometry rather than composition, have redefined the paradigm of materials science, evolving from composition dominated to architecture guided design [6–10]. By tailoring unit cell geometry and topological connectivity at multiple scales, they exhibit unconventional mechanical behaviours such as negative Poisson's ratio, quasi zero stiffness, and tunable bandgaps, endowing them with outstanding potential in energy absorption, protective structures, vibration isolation, and biomedical applications [11–16]. Nevertheless, conventional metamaterials, typically composed of a single constituent material arranged in homogeneous lattices, are limited by their intrinsic coupling between stiffness, strength, ductility, and energy dissipation [17–22]. This inherent trade-off constrains their multifunctionality and adaptability under complex, multi-field environments where strength, stability, and functional responsiveness must coexist [23–26].

To transcend these constraints, heterogeneous mechanical metamaterials have emerged as a new generation of architected systems that synergistically integrate material diversity, multiscale structuring, and functional feature fusion [27–31]. These materials break the boundaries of single phase architectures by

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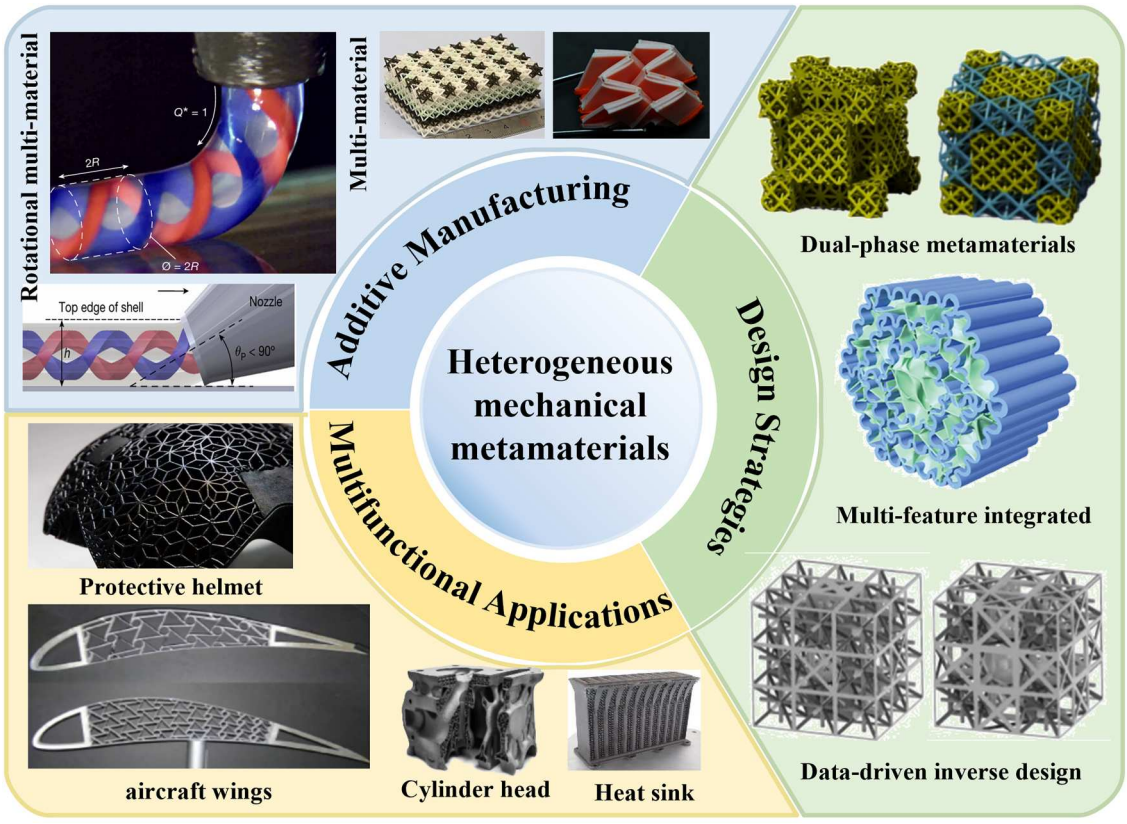


Figure 1. Schematic illustration of the research landscape of heterogeneous mechanical metamaterials, covering their structural design approaches, manufacturing processes, intrinsic structure–mechanism–property relationships, and multifunctional capabilities.

coupling different material phases, structural hierarchies, and physical fields, achieving property combinations and adaptive responses unattainable by homogeneous designs. As summarised in Figure 1, the evolution of mechanical metamaterials has entered an era where architecture, manufacturing, and function are deeply intertwined, forming a unified design paradigm that links structure–mechanism–property–performance. Building on this foundation, recent research has further refined heterogeneous design into several representative strategies that define the current landscape of the field.

As illustrated in Figure 2, current research on heterogeneous metamaterials has led to the development of six major design strategies: multiphase heterogeneous architectures, which manipulate the composition,

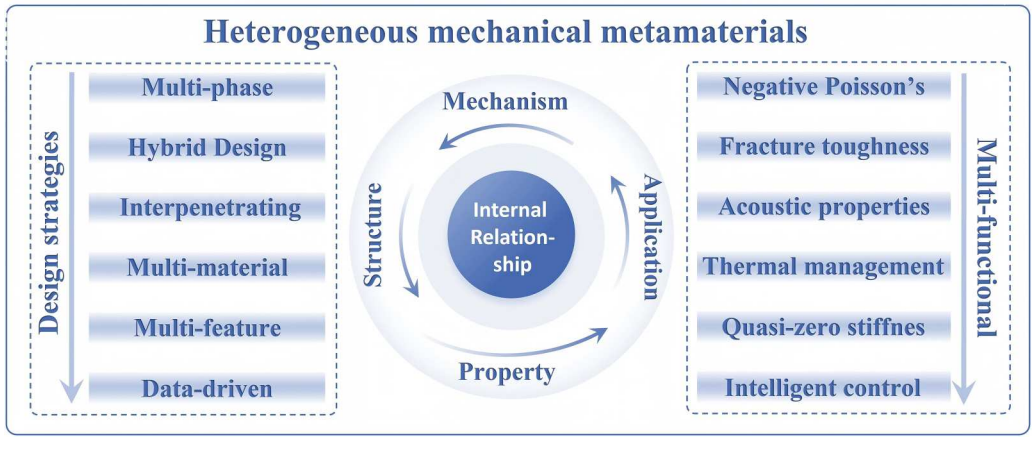


Figure 2. Framework of heterogeneous mechanical metamaterials, concept map linking six design strategies: multi phase, hybrid, interpenetrating, multi material, multi feature, and data driven, to a structure–mechanism–property–application loop, targeting multi functions including negative Poisson's ratio, fracture toughness, acoustic control, thermal management, quasi zero stiffness, and intelligent control.

proportion, morphology, and interfacial interactions of different phases to synergistically enhance strength, toughness, and energy absorption [32–37]; hybrid lattice architectures, which combine distinct lattice types or unit configurations to activate multimodal deformation mechanisms and decouple performance tradeoffs [38–42]; interpenetrating phase composites (IPCs), where multiple material networks are entangled in 3D to enable robust load transfer and energy dissipation under complex loading conditions [43–47]; multi material integration, which leverages advances in additive manufacturing to incorporate materials with contrasting mechanical or functional properties into a single framework [48–53]; multi feature fused designs, which integrate gradient configurations, bioinspired morphologies, hierarchical reinforcements, and topological innovation to unlock programmable, nonlinear, and reconfigurable behaviours [54–61]; and data driven intelligent design, which couples high throughput computation, machine learning, and generative optimisation algorithms to establish quantitative structure–property mappings, accelerate inverse design, and enable autonomous discovery of high performance heterogeneous metamaterials [62–66]. Collectively, these strategies shift the design focus from isolated geometrical tuning to architecture–composition–function coupling, thus establishing a unified framework that interlinks structure, mechanism, property, and function [67–69].

Despite remarkable progress, critical challenges remain in the development of heterogeneous mechanical metamaterials. First, a comprehensive taxonomy of heterogeneous strategies is lacking, limiting the cross comparison and generalisation of design principles. Second, the fundamental structure–mechanism–property relationships governing heterogeneous systems, particularly those involving interfacial regulation, failure evolution, and functional coupling, are not yet fully understood [70–73]. Third, the application specific design criteria for multifunctional integration remain underdeveloped, and issues related to scalable fabrication, structural robustness, and long term serviceability still hinder practical deployment [74–77]. To address these gaps, this review provides a systematic summary of the core design strategies in heterogeneous mechanical metamaterials, elucidates the characteristic unit configurations and deformation mechanisms, and discusses their effects on mechanical enhancement and multi physics performance. We further highlight representative applications across impact protection, programmable structures, adaptive electronics, thermal management, and intelligent systems [78–83]. Finally, we outline the key challenges and future directions in intelligent design, multiscale modelling, and manufacturing compatibility [84–91]. This review aims to establish a foundational reference for advancing the design and translation of heterogeneous mechanical metamaterials into next generation engineering systems.

2. Design strategy

2.1. Multi-phase heterogeneous metamaterials

Multiphase design represents one of the most representative strategies for constructing heterogeneous mechanical metamaterials. Its central principle lies in the deliberate regulation of phase type, proportion, morphology, spatial distribution, and interfacial interactions, thereby introducing cross scale strengthening and energy dissipation mechanisms that transcend the intrinsic limits of single phase lattices. As illustrated in Figure 3, a broad spectrum of multiphase architectures has been proposed, spanning from bioinspired motifs to crystal derived concepts and advanced manufacturing implementations. Figure 3a illustrates feather inspired honeycomb architectures with hierarchical interfaces, capable of steering crack propagation paths and thereby enhancing fracture toughness and energy dissipation [92,101]. Figure 3b presents a twin boundary strengthened design, where mirror symmetric unit arrangements introduce macroscopic 'twin like' boundaries, enabling controlled buckling and improved stiffness and energy absorption synergy [93]. Figure 3c shows SVPOS super cells assembled through rotation and mirroring of fundamental units; this approach allows systematic prediction and global optimisation of elastic modulus, compressive strength, and energy absorption [94]. Figure 3d depicts triclinic lattice structures with engineered twin boundaries, which effectively regulate compressive deformation modes and substantially improve energy absorption [95,102]. Figure 3e demonstrates a mortise tenon inspired modular metamaterial fabricated via 4D printing, where multiphase modules achieve lightweight, energy absorption, and repairability, overcoming the limitations of conventional metamaterials in modularity and maintainability [96]. Figure 3f illustrates a dual phase CFRP lattice sandwich, where interactive failure between matrix and reinforcement phases yields enhanced load bearing and specific energy absorption [97]. Figure 3g introduces crystal-inspired hybrid lattices, in which solid-solution-like node adjustments enable tailored stress redistribution and tunable mechanical performance [98]. Figure 3g introduces crystal inspired hybrid lattices, in which solid solution like node adjustments enable tailored

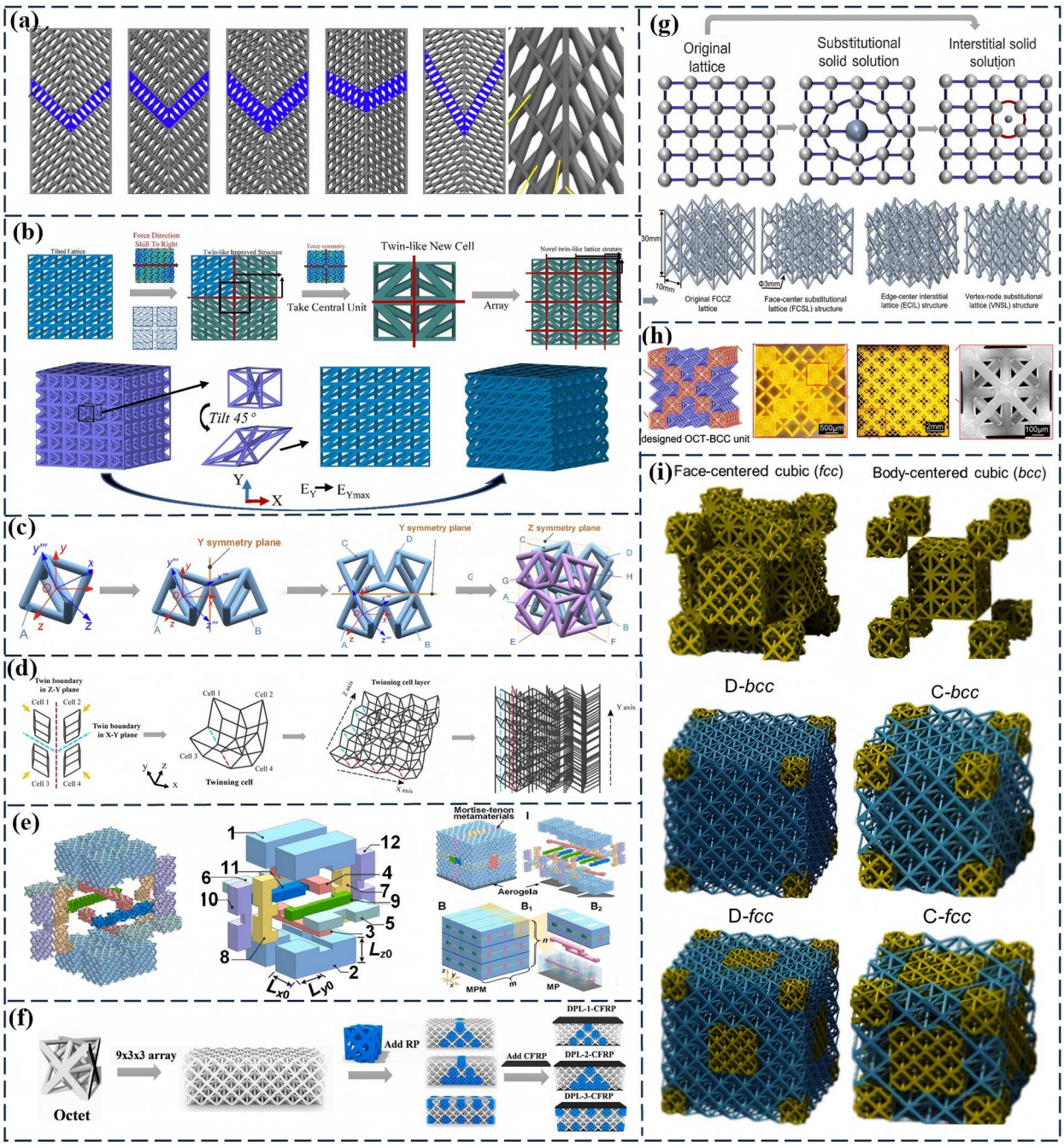


Figure 3. Representative multiphase design strategies for heterogeneous mechanical metamaterials: (a) Feather inspired hierarchical honeycombs guiding crack paths [92]; (b) Twin boundary architectures introducing macroscopic 'twin like' features [93]; (c) Rotated SVPOS assemblies for systematic performance optimisation [94]; (d) Triclinic lattices with engineered twin boundaries [95]; (e) Mortise–tenon 4D printed modular metamaterials [96]; (f) Dual phase CFRP lattice sandwiches with cooperative failure mechanisms [97]; (g) Crystal inspired solid solution lattices [98]; (h) OCT BCC dual phase microlattice [99]; (i) Precipitation strengthened dual phase lattices [100].

stress redistribution and tunable mechanical performance [98]. The high-resolution morphologies of the novel OCT BCC dual-phase micro lattice and the baseline BCC lattice at various magnifications are presented in Figure 3h, evidencing the superior mechanical properties [99]. Figure 3i shows a precipitation strengthened dual phase lattice, in which phase contrast and interfacial slip dissipation enable simultaneous enhancement of strength and toughness, with SEA up to 2.5 times higher than the single phase matrix [100,103]. Collectively, the multiphase strategies summarised in Figure 3 highlight how architectural heterogeneity governs mechanical and functional responses through identifiable physical mechanisms.

By combining bioinspired hierarchical organisation with deliberate phase contrast, these designs redistribute load across heterogeneous domains, regulate local deformation and failure modes, and activate contact- or interface-mediated dissipation. Such mechanism-level control enables the concurrent enhancement of mechanical robustness and multifunctional responses, including mechanically coupled thermal, acoustic, or field-responsive behaviours. A key advantage of geometric heterogeneity lies in its ability to activate specific deformation mechanisms rather than merely altering global stiffness or density. For example, twin-boundary like heterogeneity introduces an internal interface between architected domains, which disrupts continuous load transfer and enforces stress redistribution during loading. This interface promotes strain partitioning and sequential activation of localised bending, buckling, or contact-driven deformation on either side of the boundary, instead of synchronous collapse. As a result, deformation localisation is delayed and energy is dissipated progressively through staged instabilities, leading to enhanced stability, damage tolerance, and energy absorption even at fixed relative density.

2.2. Hybrid heterogeneous metamaterials

Hybrid heterogeneous metamaterials combine two or more distinct material phases or topological arrangements within a single structure to synergistically enhance the overall mechanical properties, energy dissipation, and multifunctional integration. These structures are designed to overcome the limitations of homogeneous metamaterials, where performance tradeoffs often occur. As shown in Figure 4a, a bending-dominated hybrid truss lattice is rationally designed to reconcile the long-standing trade-off between elastic performance and energy absorption in cellular materials. By tailoring the lattice topology, the proposed structure achieves elastic isotropy while maintaining a bending-dominated deformation mechanism, resulting in both stable nonlinear response and improved elastic stiffness. Numerical and experimental results demonstrate that, at ultralow relative densities, the lattice exhibits nearly isotropic large deformation behaviour and approaches the theoretical upper bound of Poisson's ratio for isotropic materials, while simultaneously delivering enhanced modulus, collapse strength, and specific energy absorption compared to conventional BCC truss lattices [104]. Figure 4b presents a bioinspired crossed lamellar architecture, inspired by the layered structure found in plywood, where internal orientations are carefully stacked to enhance both strength and energy absorption through hierarchical design [105]. In Figure 4c, a simple-cubic closed tubular lattice is introduced to overcome the strong anisotropy and buckling sensitivity of classical simple cubic truss lattices. By replacing slender truss members with closed tubular elements, the hybrid design suppresses premature buckling and shear instability while reducing loading-direction dependence. Enabled by microscale additive manufacturing, the resulting lattice exhibits significantly higher elastic modulus, yield strength, and stable post yield energy absorption across multiple loading directions, outperforming both simple-cubic truss lattices and state-of-the-art shellular metamaterials at comparable relative density [106,113]. Figure 4d illustrates a topological hybridisation, where a woven like structure is combined with an orthogonal lattice. This design merges the compliance of textile structures with the stiffness of the lattice framework, creating a hybrid material capable of multimodal deformation responses [107,114]. Figure 4e shows a concentric and interpenetrating hybrid design where two material phases are integrated to mitigate localised failure during tensile loading, resulting in an enhancement in both stiffness and stretchability [108]. Figure 4f highlights a multimaterial hybrid design where materials with distinct properties are integrated to achieve controlled energy absorption mechanisms, particularly useful for impact protection applications [109]. Figure 4g illustrates a frame strut hybrid design, combining bending dominated frames with axial truss members. This configuration balances lightweight design with both stiffness and strength, allowing for multifunctional adaptability and improved mechanical robustness under complex loading conditions [110]. In Figure 4h-i, an interpenetrating lattice structure with complementary morphologies is presented. This design strategy integrates lattice units of distinct topologies to optimise mechanical performance by activating multiple deformation modes, thereby enhancing overall structural resilience and energy dissipation. This hybridisation at the unit cell level results in emergent mechanical properties, enabling precise control over material behaviour at the microscale, which can be scaled up to achieve highly efficient and versatile metamaterials [111,115]. Collectively, these hybrid designs emphasise the importance of integrating different topologies or materials to achieve exceptional mechanical properties, energy absorption, and multifunctionality, pushing the boundaries of traditional homogeneous metamaterials.

2.3. Interpenetrating phase composite metamaterials

Interpenetrating phase composite metamaterials (IPCs) are a class of heterogeneous materials in which two or more continuous phases coexist within the same

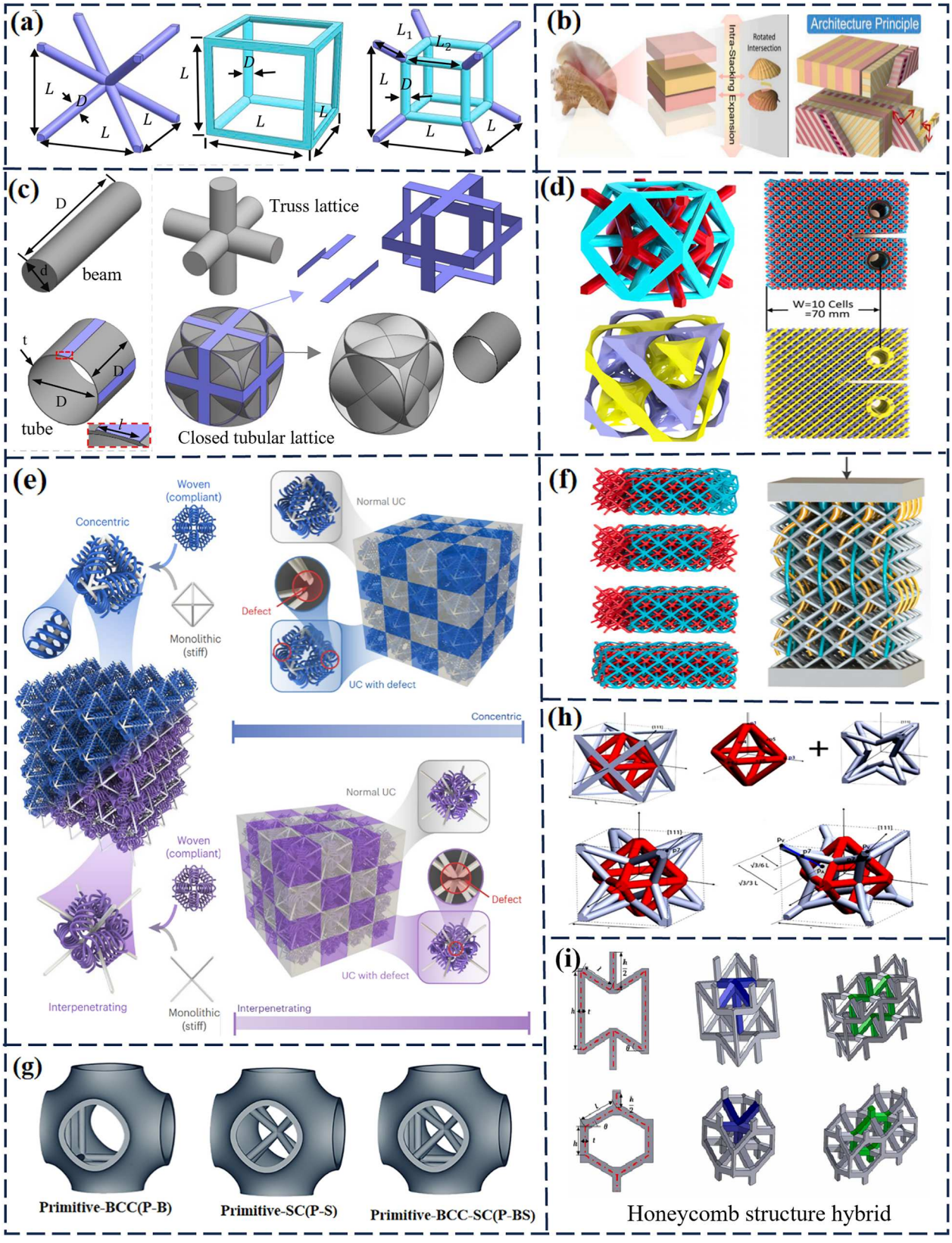


Figure 4. Representative hybrid design strategies for heterogeneous mechanical metamaterials. (a) Hybrid bending-dominated lattice enabling elastic isotropy [104]; (b) Bioinspired crossed lamellar architecture with hierarchical stacking [105]; (c) Hybrid tubular–truss lattice with reduced anisotropy and stable post-yield response [106]; (d) Co centric and interpenetrating hybrid designs [107]; (e) Topological hybridisation of woven like and orthogonal lattices [108]; (f) Multimaterial hybrid design [109]; (g) Frame strut hybrid design combining bending dominated frames and axial truss members [110]; (h i) Hybridised unit cells design logic [111,112].

microstructure, enabling unique mechanical and functional performance that surpasses what single phase materials can offer. By interlocking distinct material networks, IPCs combine the structural benefits of multiple phases to enhance load bearing capacity, energy absorption, and damping under complex loading conditions. As shown in Figure 5, IPCs leverage the interfacial interactions between different material phases to create multifunctional designs with superior properties. Figure 5a presents an aperiodic truss lattice inspired by the two-dimensional 'Einstein' monotile, which is extended into three-dimensional truss units. These units exhibit unique spatial patterns based on angle and tiling methods, and the surrounding air spaces are filled with solid materials, forming IPC metamaterials with enhanced mechanical properties [116]. Figure 5b shows the fabrication of bicontinuous IPCs based on the Schwarz Diamond surface, produced using FDM 3D printing. This configuration improves the mechanical robustness and energy absorption by embedding multiple phases with continuous connectivity [117]. In Figure 5c, the damping behaviour of Al epoxy IPCs is tested under both static and dynamic loading conditions, demonstrating the excellent energy dissipation and toughness achieved through hybridisation [118]. Figure 5d illustrates a multi-sheet lattice structure based on the Gyroid surface, designed using TC4 laser powder bed fusion (LPBF) and epoxy resin to create an interpenetrating phase composite. This structure exhibits superior damping and stability due to the multiphase composition [119]. As noted in Figure 5e, a new concept called hierarchical IPCs (HIPCs) was introduced. Polymer was infused into 3D printed hierarchical ceramic lattices [120]. In Figure 5f, a method was developed to make Al2O3/Al composites with bicontinuous networks. It aims to solve problems like broken reinforcement and stress concentration in normal Al composites. A bio inspired gyroid structure was used as the ceramic scaffold. This improves energy absorption and plastic deformation [121]. Figure 5g presents a hybrid design combining two distinct materials with high contrast in mechanical properties. This configuration features a bimaterial lattice structure, where the contrast between the materials enables enhanced mechanical performance, particularly in terms of load distribution and fracture resistance under tensile stress. The interpenetrating phases within the lattice structure synergistically work together to improve the energy absorption capacity [122]. Figure 5h presents an AlSi10Mg NiTi lattice structure IPC, fabricated using additive manufacturing and vacuum infiltration. The IPC demonstrates an improvement of 94% in damping capacity and an increase of 100% in compressive strength compared to AlSi10Mg, attributed to improved interface damping and nanoscale grain strengthening. [123,124]. Collectively, IPCs exemplify how heterogeneous phase integration can provide high strength, energy dissipation, and multifunctional performance, offering broad potential for applications in lightweight engineering, impact resistance, and damping technologies.

Although hybrid architectures and interpenetrating-phase composites both exploit structural heterogeneity, their underlying load-transfer and failure mechanisms are fundamentally different. In hybrid architectures, mechanical functionality is achieved by combining discrete structural subsystems with distinct roles, such as stiff load-bearing frameworks and compliant or dissipative components. Load transfer is primarily mediated through designed geometric connections, and deformation is typically activated in a hierarchical or sequential manner. As a result, failure tends to localise within specific substructures through yielding, buckling, or fracture, enabling staged damage evolution and controllable collapse pathways. By contrast, interpenetrating-phase composites consist of two or more continuous phases that form intertwined load-bearing networks throughout the material volume. Load sharing occurs at the phase level rather than through discrete joints, leading to distributed stress transfer. Their failure behaviour is therefore dominated by interfacial debonding, phase bridging, and crack deflection or arrest, with damage developing in a more spatially distributed manner rather than through discrete structural collapse.

2.4. Multimaterial mechanical metamaterials

Multimaterial heterogeneous metamaterials integrate two or more materials with distinct properties to achieve superior mechanical performance that a single phase material cannot provide. By combining different material phases, these metamaterials can optimise mechanical properties. As shown in Figure 6, multimaterial designs leverage the synergistic effects between different materials to create high performance, multifunctional structures. Figure 6a features a large scale octet truss made from two materials. It was fabricated by centrifugal multimaterial 3D printing. This technique allows efficient spatial separation and integration of materials within one lattice. Therefore, it demonstrates good scalability and control of heterogeneity [125]. Figure 6b shows a gyroid based structure with Ti-6Al-4V, Cu, and Al-Cu-Mg phases. A Cu interlayer improves interfacial bonding. It also enables smoother thermal and mechanical transition between materials [126,127]. Figure 6c illustrates rotational multimaterial printing of

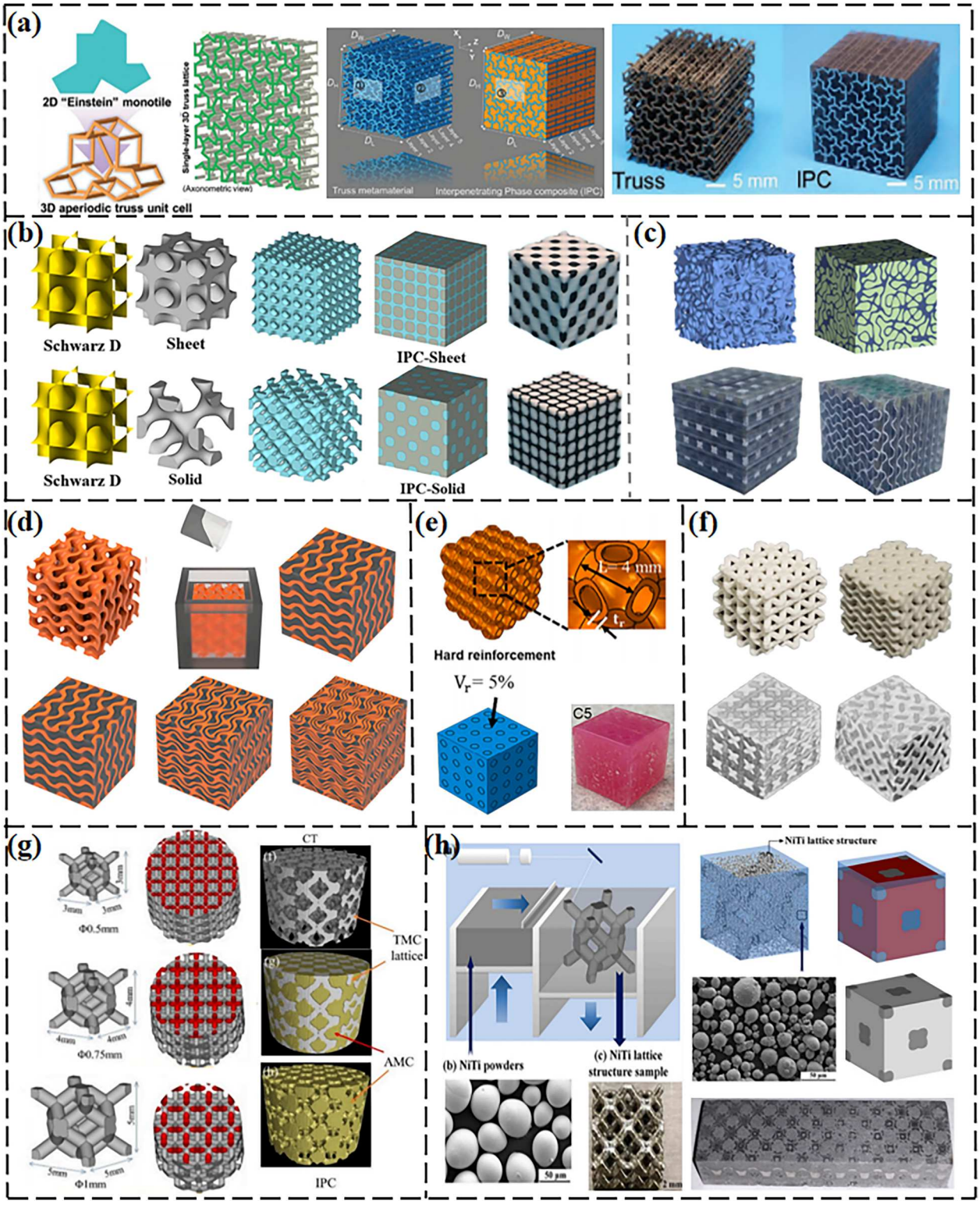


Figure 5. Design strategies for interpenetrating phase composite (IPC) metamaterials. (a) Aperiodic truss lattice inspired by the 'Einstein' monotile extended into 3D truss units with solid material filling [116]; (b) Bicontinuous IPCs based on the Schwarz Diamond surface [117]; (c) Damping behaviour of Al epoxy IPCs [118]; (d) Multi-sheet lattice based on Gyroid surface formed using LPBF with epoxy resin [119]; (e) Density controlled lattice fabrication via PolyJet printing [120]; (f) Al2O3 ceramic lattices achieve programmable stiffness via geometric tuning [121]; (g) Hybrid bi material IPC lattice structure [122]; (h) High damping NiTi/AlSi10Mg dual metal IPCs [123].

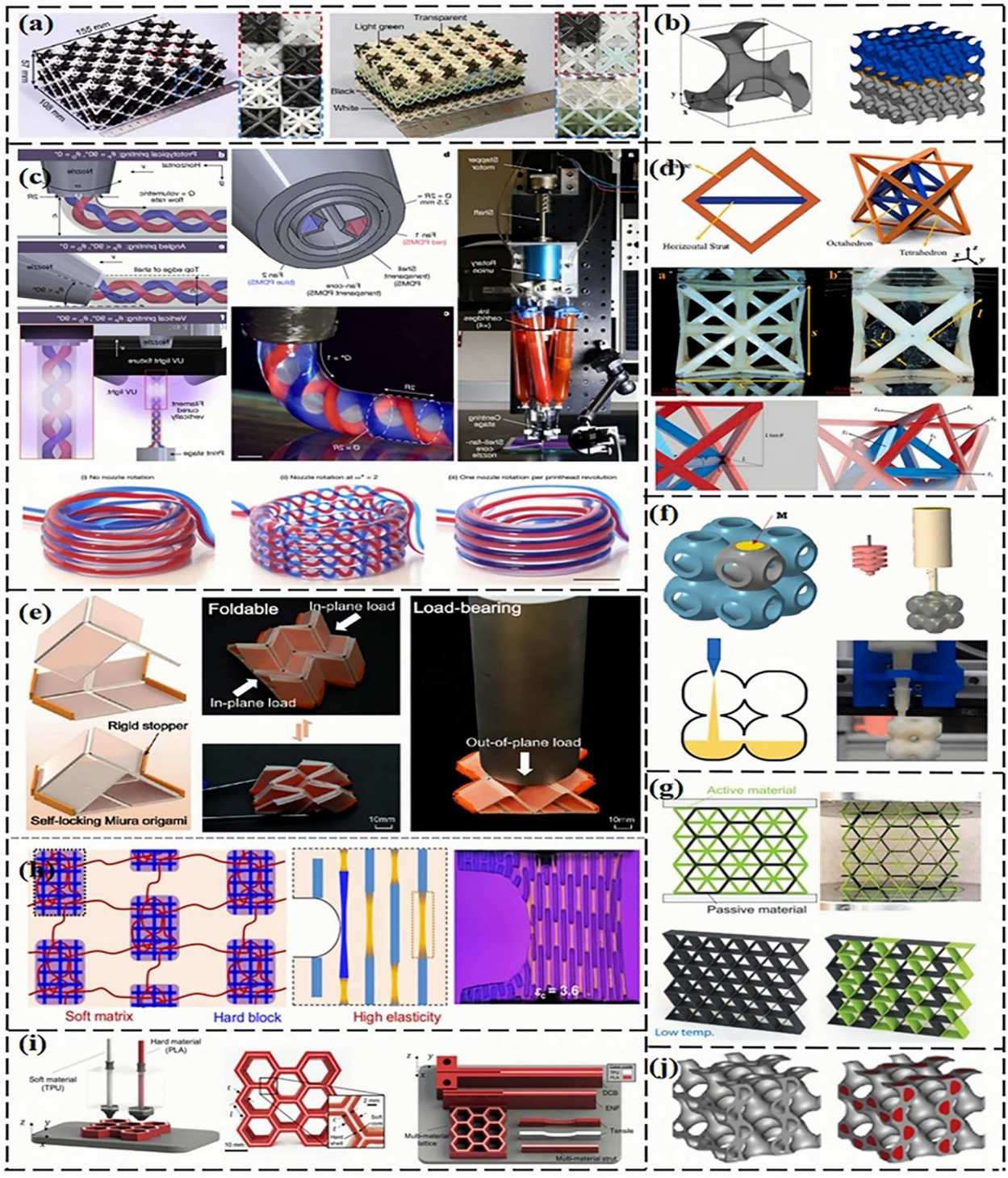


Figure 6. Design and fabrication of multimaterial metamaterials. (a) Centrifugal 3D printing of large scale dual material octet trusses [125]; (b) Additively manufactured Ti–6Al–4 V/Al–Cu–Mg gyroid with Cu interlayer [126]; (c) Rotational 3D printing of subvoxel controlled PDMS filaments [128]; (d) Tunable mechanics via diamond/octet truss multimaterial lattices [129]; (e) Multimaterial printed self locking thick panel origami [131]; (f)Foam-filled closed-cell lattices [132]; (g) Thermally programmable active passive triangular lattices [133]; (h) Tough topoarchitected rigid soft polymer networks [135]; (i) FFF printed honeycombs with adjustable wall geometry [136]; (j) Bioinspired co continuous gyroids with tunable density [137].

architected PDMS filaments. Precise nozzle rotation creates helical multimaterial filaments. This method allows tunable mechanical anisotropy across multiple scales [128]. As shown in Figure 6d, a diamond octet truss is modelled with bi material unit cells. Different materials are assigned to specific struts. This allows

systematic study of stiffness, strength, and Poisson's ratio at the lattice level [129,130]. Figure 6e presents a self-locking thick panel origami metamaterial. It is made from multimaterial printed components. The Miura origami design offers in plane foldability and high out of plane load capacity through geometric interlocking [131]. As visualised in Figure 6f, it shows closed cell lattices filled with lightweight foams. A hybrid printing and infilling method ensures uniform foam distribution. This enhances energy absorption and reduces density [132]. Figure 6g demonstrates a triangular multimaterial lattice made with active and passive thermomechanical materials. These structures show programmable mechanical behaviour under temperature changes. They are promising for adaptive applications [133,134]. Figure 6h displays a staggered topologically entangled polymer network (TPN). Rigid polymer blocks are embedded in a soft matrix. This design balances high stretchability and toughness via network interpenetration [135]. Figure 6i depicts a honeycomb lattice made by multimaterial FFF printing. By changing cell wall composition and geometry, properties like energy absorption and load capacity can be precisely adjusted [136,138]. Finally, Figure 6j shows a co-continuous gyroidal composite inspired by natural cellular structures. Boolean operations help create periodic structures with tunable composition and density. This leads to multifunctional mechanical performance [137].

In multi-material fusion, sharp material interfaces represent the most defect-sensitive heterogeneity. Mismatches in rheology, curing behaviour, or thermal response commonly lead to interfacial porosity, weak bonding, and material cross-contamination, which limit load transfer and trigger early interfacial failure. To mitigate these effects, recent studies employ graded transition layers to smooth stiffness and property jumps, interface-aware toolpath planning to enhance bonding continuity, and post-processing treatments to reinforce interfacial adhesion [138,139]. Such strategies are essential for achieving reliable mechanical performance in multi-material heterogeneous metamaterials. Together, these multimaterial heterogeneous metamaterials demonstrate how heterogeneous phase integration can be used to enhance mechanical, thermal, and energy dissipation properties, making them ideal candidates for a wide range of applications, from lightweight engineering to impact resistant materials.

2.5. Multifeature integrated bioinspired heterogeneous metamaterials

In recent years, research on mechanical metamaterials has increasingly shifted from optimising single functions to integrating multiple structural features within heterogeneous material systems. This transition focuses on overcoming the inherent trade-offs between mechanical properties, aiming for a balanced improvement across these features. Traditional structures often rely on a single deformation mode, which limits their overall performance. In contrast, bioinspired multifeature integrated architectures, which combine different material phases and structural types, activate multiple deformation mechanisms across different length scales, enabling the design of more versatile, high performance materials. As illustrated in Figure 7a, a modified FCC lattice (MFCC) is inspired by the skeleton of deep sea glass sponges. It replaces the traditional single diagonal struts with double diagonal ones. A hierarchical circular support system is added. Together with the dual diagonal bracing, this forms the MHCFCC structure. The double diagonals provide strong support. Meanwhile, the circular units promote progressive buckling. This combination leads to high strength, energy absorption, and good structural stability [140]. Figure 7b shows a bioinspired composite that merges S shaped cuttlebone channels, spider web patterns, and the porous structure of pomelo peel. These elements are integrated into a flexible multilayer material called S web foam. It shows excellent damping, energy storage, and cushioning. When used in a slipper heel, it improves both comfort and safety [141,145]. Figure 7c depicts a checkerboard lattice improved by a gradient hierarchical design. It solves the trade-off between mechanical strength and controllability. Inspired by biological tissues, the design introduces axial gradients in unit shape, wall thickness, and density. These features activate a multistage, coordinated crushing process [142]. According to Figure 7d, a dual phase cylindrical lattice tube is created. It takes inspiration from the microstructures in the raptorial appendages of the peacock mantis shrimp. The structure includes matrix (MP) and reinforcement (RP) phases. The RP phase is carefully arranged to improve energy absorption. This tube is designed by a mapping method. It is then produced using additive manufacturing and tested under quasi static compression. The result is a material that combines high strength and high energy dissipation [143,146]. Finally, Figure 7e presents a metamaterial with both adaptability and mechanical strength. It includes a reinforced exoskeleton and an internal sponge like rib network. The structure is divided into functional zones. This allows better control of deformation and improves mechanical response. A rotating polygonal frame inside the design creates multiple plateau stress stages [144].

In bioinspired heterogeneous metamaterials, performance originates from transferable functional

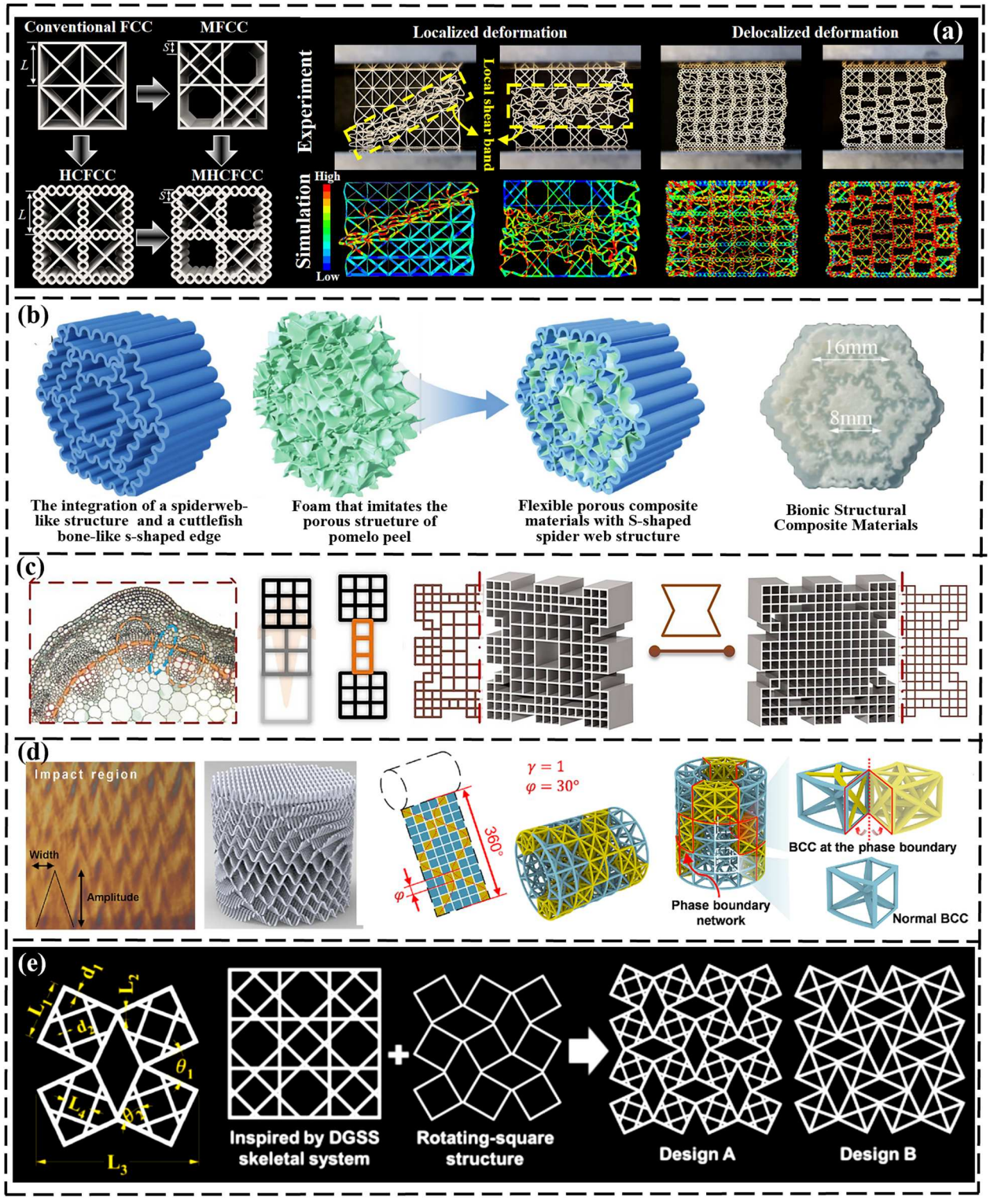


Figure 7. Design strategies for multifeature integrated heterogeneous metamaterials. (a) Modified FCC lattice with double diagonal struts and hierarchical circular support system, integrating multiple structural features for enhanced mechanical performance [140]; (b) S web foam combining S shaped cuttlebone channels, spider web patterns, and porous layers [141], (c) Checkerboard lattice with a gradient hierarchical design, enabling staged deformation and multiphase energy dissipation [142]; (d) Dual phase cylindrical lattice inspired by mimics mantis shrimp arm, combining bioinspired design and dual phase structure [143]; (e) Hybrid design combining bioinspired double diagonal bracing and rotating polygonal frame structures [144].

principles rather than geometric resemblance. Biological materials regulate load and damage through graded stiffness, distributed load transfer, and sequential activation of deformation mechanisms. These principles are implemented in multifeature integrated metamaterials via architected heterogeneity, such as stiffness or geometry gradients to redistribute stress, multistage buckling or snap-through to control stability and energy dissipation, and distributed deformation modes that delay localisation and failure. Adaptive biological functions, including stimulus triggered reconfiguration and recovery, are translated into tunable stability landscapes using material mismatch or contact-mediated interactions, enabling reversible switching of mechanical response without direct morphological replication.

2.6. Data-driven metamaterial design

Data driven strategies are accelerating the discovery of heterogeneous mechanical metamaterials by coupling machine learning (ML) with architectures that deliberately mix multiple phases, features, and length scales to elicit emergent responses that single feature designs cannot achieve. As illustrated in Figure 8a, an inverse design pipeline that integrates deep neural networks with genetic algorithms identifies truss lattices with superior buckling resistance by searching heterogeneous topological motifs assembled from diverse building blocks, thus linking geometry topology heterogeneity to stability gains [147]. Figure 8b generates three dimensional auxetic metamaterials from implicit TPMS expressions and then heterogenises function via 3D printing and electroless Ni plating, jointly tuning stiffness, strength, and conductivity while preserving auxeticity; the panel contrasts design encodings (parameters, voxels, meshes) that control how geometric heterogeneity is conveyed to ML models [148,154]. Figure 8g presents a growth-based, data-driven topology optimisation framework that enables the generation of irregular metastructures through controlled yet stochastic assembly of predefined building blocks on unstructured graphs. By simultaneously optimising block selection and geometric transformations (scaling, skew, and rotation), the method explores a large, heterogeneous design space and produces metastructures with optimised mechanical functionalities, uniform feature sizes compatible with fabrication, and natural conformity to complex domain boundaries, making them suitable as mechanical infills [149]. Figure 8d employs a neural accelerated evolutionary strategy in which a surrogate network serves as a fast forward solver and, when combined with evolutionary search, inverts for unit cell geometries that realise targeted nonlinear responses arising from heterogeneous deformation pathways [150]. Figure 8e demonstrates rapid inverse design directly from prescribed stress–strain trajectories while accounting for manufacturing errors; by superimposing gradients in unit cell type and strut radius ratios, it composes spatially heterogeneous composites whose macroscopic response closely tracks the target curve [151]. In Figure 8f, a multi objective NSGA II framework discovers lattices that are simultaneously isotropic and auxetic, even though none of the eight base cells individually possess these traits, by optimising their heterogeneous spatial arrangement under symmetry constraints [152]. In Figure 8c, a generative computational framework is employed to design irregular, non-periodic architected materials by optimising the spatial distribution of a limited set of basic building blocks. The resulting disordered microstructures exhibit spatially varying mechanical properties that precisely modulate stress towards prescribed target levels across multiple control regions and loading scenarios, reflecting the stress-regulation capability of natural materials. The stress-programmable architectures are experimentally validated using 3D-printed specimens and further demonstrated in orthopedic femur restoration, where controlled stress redistribution is used to promote bone repair [153].

Despite the success of machine-learning-assisted design in mechanical metamaterials, model interpretability remains a major limitation. Most frameworks operate as black-box predictors and provide limited insight into how geometry, material heterogeneity, or interfaces govern deformation, stability, and failure. This issue is particularly severe in heterogeneous metamaterials, where localised interface effects, contact activation, and nonlinear damage evolution dominate the response and are difficult to extract from latent representations. Physics-informed learning, geometry-based feature attribution, and hybrid learning–mechanism models have shown progress [154], but transferable structure–mechanism–property relationships are still lacking. A related challenge is the gap between computationally optimal designs and their manufacturability or mechanical reliability. Learning pipelines often assume idealised geometry and defect-free fabrication, resulting in architectures with ultra-thin members, sharp multimaterial interfaces, or highly localised stress paths that are sensitive to printing defects and service loading. Addressing this gap requires embedding manufacturability constraints (e.g. minimum feature size, self-support limits, interface resolution) and reliability-oriented criteria (e.g. stress localisation and interface failure) directly into learning and optimisation, together with defect-

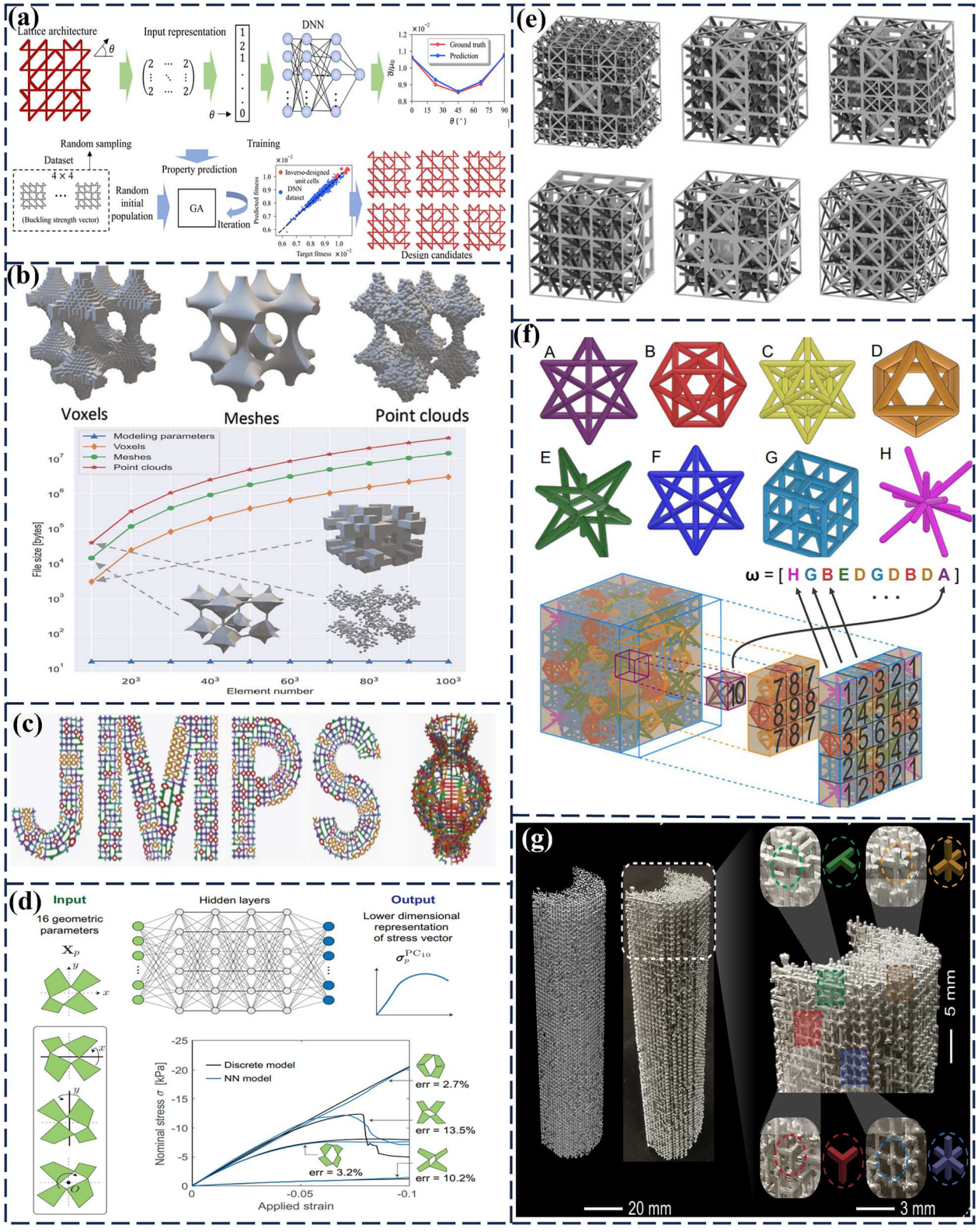


Figure 8. Machine learning–driven design of heterogeneous mechanical metamaterials. (a) DNN + GA inverse design of buckling resistant lattices [147]; (b) Auxetic representations: parameters/voxels/meshes/point clouds [148]; (c) Growth-based topology optimisation for irregular metastructures [149]; (d) NN stress–strain surrogate vs. physics models [150]; (e) Inverse design from target curves [151]; (f) NSGA II modular lattices from eight base cells [152]; (g) Stress-programmable irregular metamaterials via generative block assembly [153].

aware training and physics-based screening prior to experimental validation [155].

Taken together, the six heterogeneous design strategies discussed above exhibit distinct mechanism-driven advantages as well as clear practical boundaries. Multiphase design primarily exploits stiffness and strength contrast among constituent phases to achieve load partitioning and damage mitigation, making it effective for enhancing robustness and energy absorption, while its performance is often constrained by interface integrity, phase continuity, and manufacturing-induced defects. Hybrid architectures, which combine different topological motifs within a single structure, enable controlled deformation sequencing and stability regulation, but their effectiveness is sensitive to geometric compatibility and fabrication tolerance, limiting scalability. Interpenetrating composites rely on continuous load transfer between mutually interpenetrating phases to improve damage tolerance and multifunctionality; however, their application is frequently restricted by infiltration feasibility, curing-induced residual stresses, and density penalties. Multi-material fusion provides a powerful route to programme anisotropy and spatially varying functionality through material distribution, yet its practical deployment is challenged by material compatibility, bonding reliability, and process integration. Multi-feature integration, which couples hierarchy, contact nonlinearity, anisotropy, and other architectural features, can substantially expand achievable performance envelopes, but often at the cost of increased design complexity and reduced interpretability, complicating optimisation and transfer across scales. Data-driven approaches offer an efficient means to explore the high-dimensional design space of heterogeneous metamaterials and identify non-intuitive architectures, while their reliability remains bounded by dataset quality, generalisation capability, and the difficulty of embedding failure physics and manufacturability constraints. This comparative perspective highlights that each strategy occupies a distinct regime of applicability, and effective design requires balancing targeted performance gains against structural complexity and practical constraints.

3. Structure–mechanism–property relationships

Building on the heterogeneous design strategies outlined in Chapter 2, this section examines how these architected concepts govern mechanical performance and multifunctional properties. Particular emphasis is placed on negative Poisson's ratio effects, fracture toughness, acoustic and thermal characteristics, as well as intelligent responses and bistable behaviours, highlighting the underlying structure–property relationships enabled by architectural heterogeneity.

3.1. Negative Poisson's ratio effect

Negative Poisson's ratio (NPR) metamaterials provide a canonical route to heterogeneous structure–mechanism–property coupling, where spatially varied motifs, hybrid unit features, gradients, and local mass additions are orchestrated to elicit lateral expansion and multifunctionality that homogeneous media cannot achieve. Across Figure 9, NPR is not treated as a single geometry but as a heterogeneous design space in which auxetic response, stiffness, isotropy, damping, bandgaps, and energy absorption are cotuned via multifeature integration. In Figure 9a, a sign switchable Poison's ratio arises from a heterogeneous fusion of modified hexagonal and concave units, enabling programmable stiffness and deformation modes [156]. Figure 9b establishes directional NPR and enhanced energy absorption by combining bending dominated and buckling induced mechanisms, demonstrating how mechanism level heterogeneity empowers directional protection and dissipation [157]. In Figure 9c, a recurve bow shaped re-entrant honeycomb is proposed to stabilise negative Poisson's ratio behaviour under large deformation by introducing curved ligaments into the classical re-entrant topology. This geometric heterogeneity couples elastic bending with nodal rotation, mitigating stress concentration at sharp corners and enabling the negative Poisson's ratio to persist over a substantially extended strain range, while improving tensile–compressive consistency and stress uniformity compared to conventional re-entrant honeycombs [158]. Figure 9d presents a modularly assembled auxetic honeycomb that realises negative Poisson's ratio through misaligned interlocking of corrugated plates with different functional roles. By transitioning from an integrated microstructure to a heterogeneous assembly architecture, the design preserves auxetic deformation while enhancing manufacturability, structural stability, and scalability. The resulting negative Poisson's ratio system achieves low crush stress and stable energy absorption, highlighting how assembly-induced heterogeneity can support NPR behaviour under severe loading [159]. In Figure 9e, negative Poisson's ratio deformation is combined with strain-dependent stiffness modulation to achieve multifunctional acoustic performance. A variable stiffness factor is embedded into perforated auxetic plates, enabling real-time tuning of bandgap characteristics as the negative Poisson's ratio structure deforms. This coupling between NPR-induced

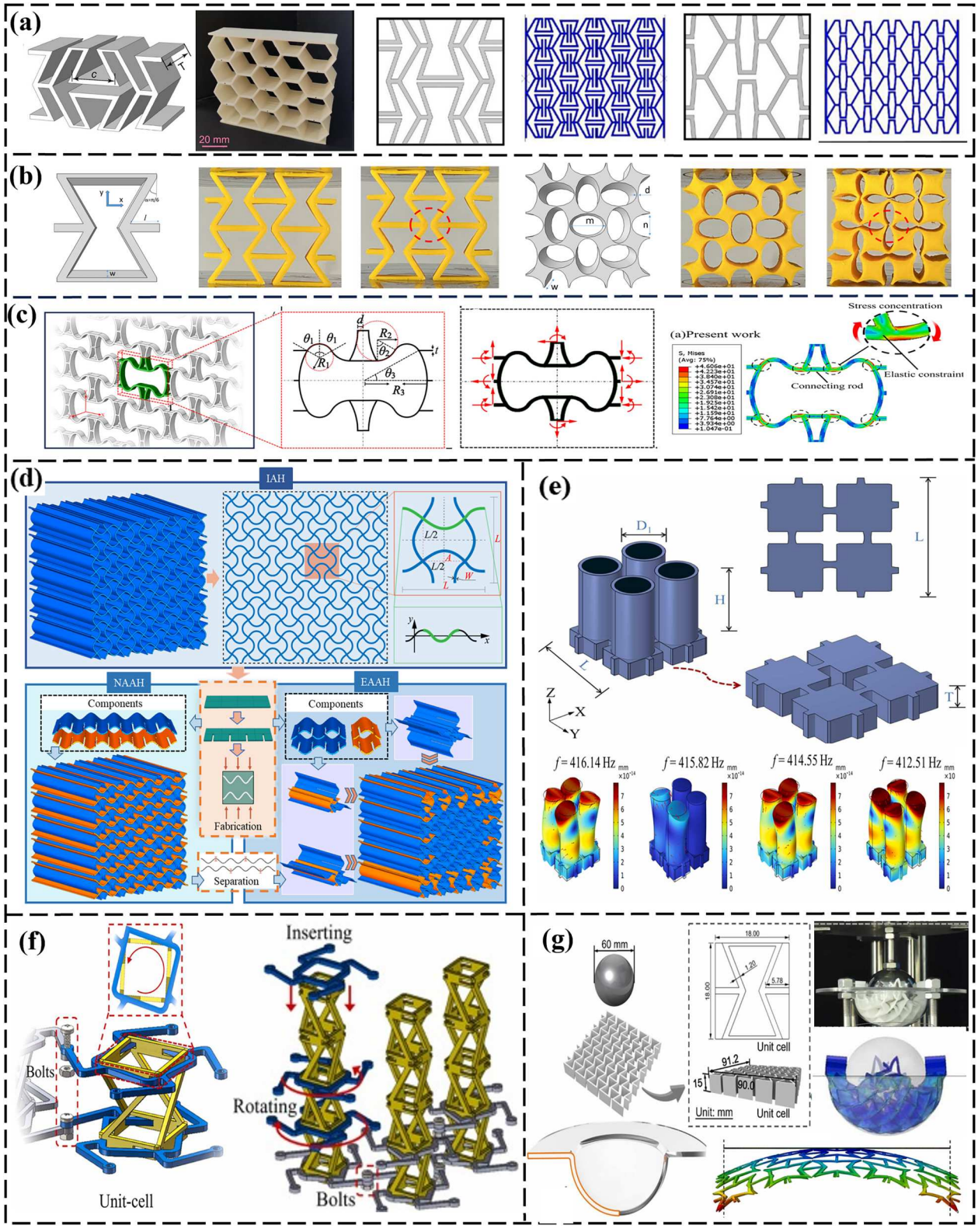


Figure 9. Structural variations in 2D/3D negative Poisson's ratio heterogeneous metamaterials, (a) PR sign switching PRSSM [156]; (b) Buckling/bending tensile dilation lattices [157]; (c) Negative Poisson's ratio honeycomb with curved ligaments for stabilised auxetic deformation [158]; (d) Modularly assembled heterogeneous negative Poisson's ratio honeycomb based on misaligned interlocking [159]; (e) Negative Poisson's ratio acoustic metamaterial with strain-tunable bandgaps [160]; (f) Assembled heterogeneous metamaterial with coupled compression–twisting response [161]; (g) Topology-dependent metamaterials for synclastic curvature and conformability [162].

kinematics and acoustic bandgap evolution demonstrates how auxetic architectures can serve as adaptive platforms for multi-physics property integration [160]. Figure 9f introduces an assembled auxetic meta-structure that exhibits negative Poisson's ratio through coordinated compression–twisting deformation of discretely manufactured components. Structural heterogeneity at the component and connection levels generates stable interlocking constraints, allowing the NPR structure to sustain large deformation with a stable stress plateau. The assembly-based strategy enables tunable mechanical response and energy absorption while maintaining negative Poisson's ratio behaviour using low-cost conventional fabrication methods [161]. Finally, Figure 9g systematically examines how different negative Poisson's ratio architectures accommodate synclastic curvature, linking NPR-induced lateral expansion to surface conformability and contact stress distribution. Comparative tests reveal that re-entrant structures offer superior conformity and reduced contact stress, whereas other auxetic topologies suffer from local buckling, underscoring how topology-dependent heterogeneity governs the practical usability of negative Poisson's ratio metamaterials in ergonomic and engineering applications [162]. Collectively, these heterogeneous negative Poisson's ratio metamaterials demonstrate that stable auxetic behaviour is not dictated by a single re-entrant geometry, but emerges from the deliberate integration of curved ligaments, modular assembly, multi-physics coupling, and topology-dependent deformation mechanisms [163-165]. By exploiting heterogeneity at the geometric, architectural, and functional levels, NPR metamaterials achieve enhanced strain-range stability, programmable mechanical response, and application-oriented performance [166].

3.2. Fracture toughness

Fracture toughness in heterogeneous mechanical metamaterials emerges from deliberate spatial, material, and topological heterogeneity that redirects, blunts, or dissipates cracks across multiple scales, mimicking nature's dual network tissues, graded shells, and cellular architectures.

To facilitate comparison across different heterogeneous architectures, fracture related performance is commonly evaluated using a small set of normalised indicators. Representative metrics include specific fracture energy (fracture energy normalised by density), crack initiation strain, and crack deflection or arrest efficiency, which quantify how architectural heterogeneity redistributes stress and interrupts crack propagation.

As shown in Figure 10a, damage-programmable (DP) materials encode spatially heterogeneous sacrificial units to orchestrate crack tip interaction, shielding, and bridging, reprogramming paths and elevating R curve behaviour through staged energy dissipation [168,172]. Figure 10b contrasts an octet truss with a Schwarz P shell, revealing that morphological heterogeneity (truss vs. shell micro mechanisms) redistributes stress and plastic zones; DIC informed FE models and tests show Schwarz P shells dissipate more energy and resist crack advance more effectively than octets under otherwise comparable conditions [169]. Figure 10c shows a dual bond metamaterial integrating weak and strong bonds. By guiding the sequential fracture of weak bonds and the formation of plastic hinges, stress is ingeniously redistributed throughout the structure, achieving full field energy dissipation and unprecedented toughening effects. [170]. Figure 10d maps how geometric and thickness heterogeneity tunes stress triaxiality and failure modes in 3D lattices: XCT imaging and force–displacement data show octet lattices transition from crack-driven failure to elastic buckling as crack size/thickness varies, establishing a heterogeneity–triaxiality–toughness linkage crucial for robust 3D deformables [171]. Collectively, Figure 10 shows that fracture resistance rises when heterogeneity is treated as a design variable, via programmed damage units, dual networks, graded geometry/thickness, and bi-material interfaces, rather than an imperfection to be minimised [173].

3.3. Acoustic properties

Acoustic metamaterials achieve exceptional wave control when heterogeneity, in phase composition, geometry, coupling pathway, and spatial gradation, is treated as the primary design lever rather than an afterthought. For acoustic performance, heterogeneous metamaterial architectures are typically compared using normalised descriptors rather than absolute frequencies. Commonly adopted indicators include normalised bandgap width (bandgap width normalised by the central frequency), bandgap-to-density ratio, and attenuation level within the effective bandgap range. These metrics enable direct comparison of how different geometric and material heterogeneities broaden bandgaps, shift operating frequencies, or enhance wave attenuation efficiency under comparable mass and length-scale constraints.

As shown in Figure 11a, an interwoven bi-phase micro lattice reconstructs open air domains into distributed Helmholtz-like cavities while a load-bearing skeleton carries stress, enabling simultaneous sound absorption and mechanical support; the acoustic

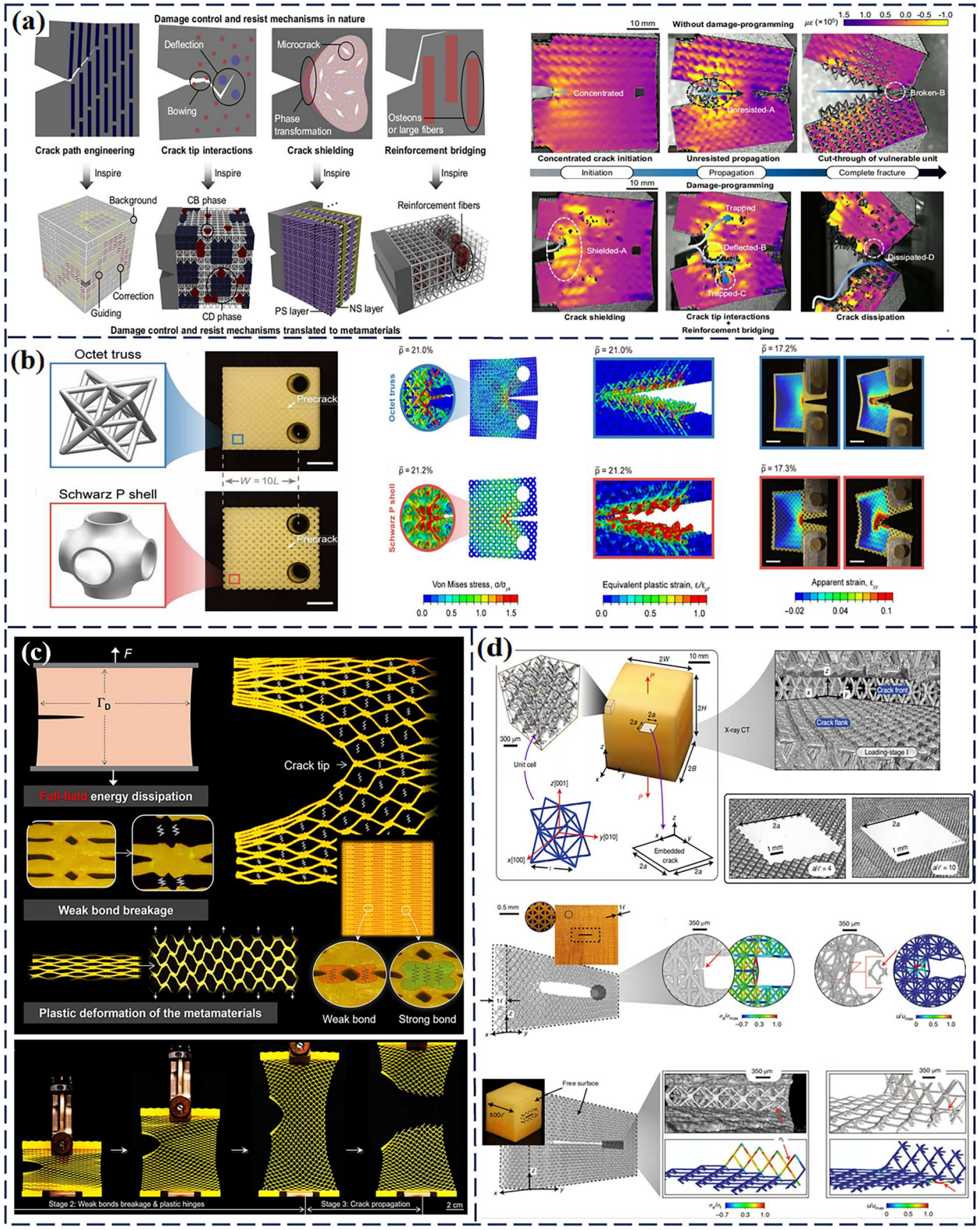


Figure 10. Bio-inspired and architected heterogeneous metamaterials for damage tolerance and fracture resistance. (a) Nature-inspired, damage-programmable crack control (shielding/bridging/path deflection) [168]; (b) Fracture behaviour in contrasting topologies (octet truss vs. Schwarz P shell) [169]; (c) Fracture behaviour of double-bonded metamaterials [170]; (d) 3D crack propagation in lattices analysed by XCT and FE; thickness/crack size effects on triaxiality and failure mode [171].

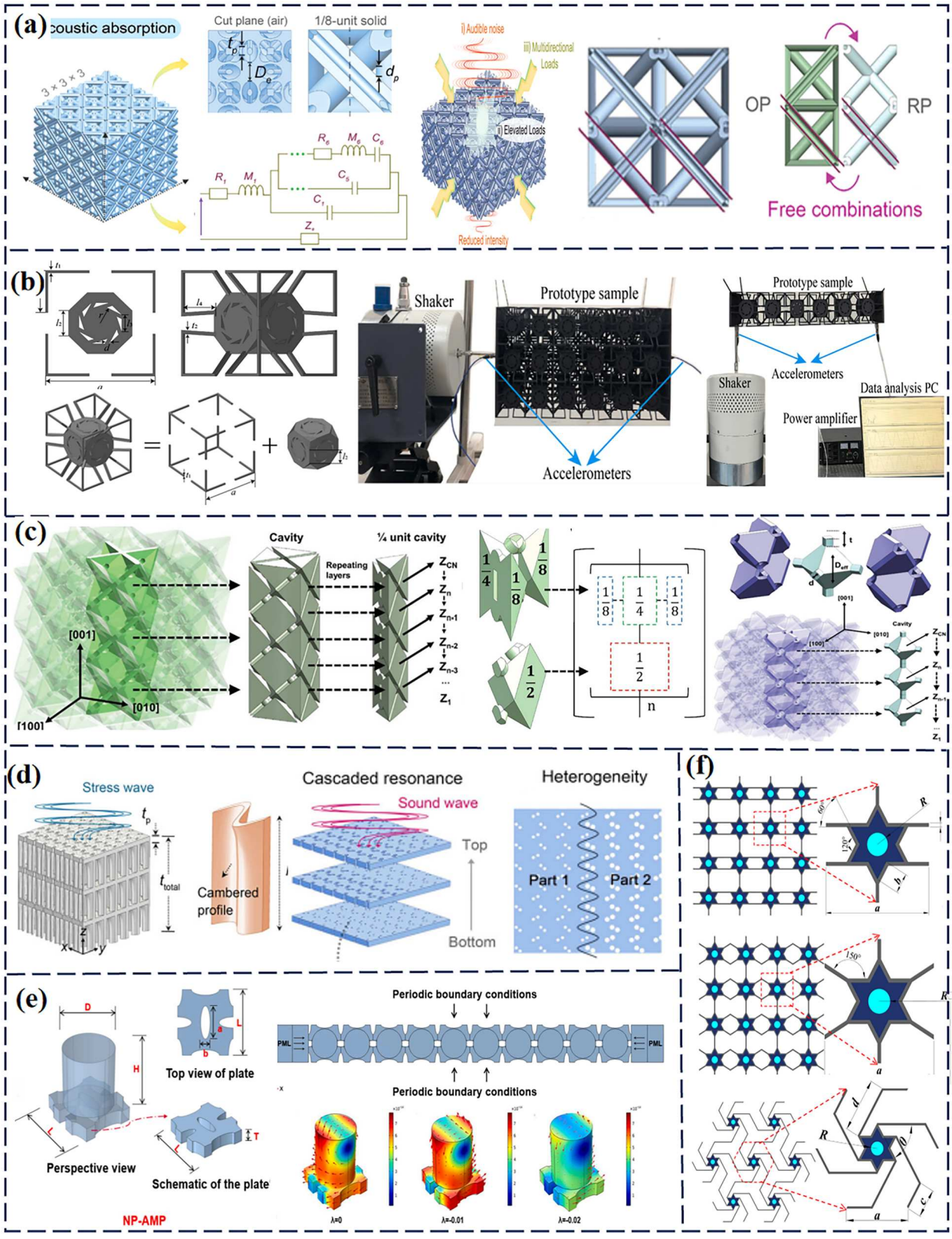


Figure 11. Multifunctional heterogeneous acoustic/mechanical metamaterials, (a) Dual phase interwoven micro lattice for concurrent load bearing and absorption [174]; (b) Unit cell prototype test workflow validating broadband attenuation [175]; (c) Gradient acoustic lattices with impedance/geometry heterogeneity [176]; (d) Functional decoupling of resonant plate and load bearing frame [177]; (e) NP AMP plate FE model enabling low frequency bandgap tuning [178]; (f) Star unit 2D lattices with rectangular/orthogonal/triangular outer frames [179].

channel is circuit modelable and tunable across a broad bi-phase geometric space [174]. Figure 11b hybridises a chiral plate (as a locally resonant mass) with a surrounding rod frame (as a support), creating a rod–chiral plate heterogeneous resonator that delivers broadband vibration attenuation, validated on dual testbeds [175]. In Figure 11c, two architected plate lattices, FCC Plate and Fluorite Plate, encode impedance heterogeneity via distinct cavity–pore networks: the former behaves as serially cascaded Helmholtz units, whereas the latter forms a mixed parallel series complex network, jointly enabling tailored absorption spectra [176]. Figure 11d exemplifies functional decoupling through a cephalopod bone-inspired design: a porous resonant plate independently sets acoustic response while arc-shaped unit walls (mimicking asymmetric growth) independently bear load; two pathways combined into one heterogeneous entity [177]. Figure 11e leverages a negative Poisson's ratio plate whose rotation-driven unit kinematics (shown by PBC FE analysis) downshifts bandgaps to low frequencies, illustrating how kinematic heterogeneity tunes dispersion [178]. Finally, Figure 11f uses a common star-shaped internal resonator coupled with three topologically distinct outer link networks (rectangular, orthogonal, triangular), demonstrating that outer frame heterogeneity steers Bragg, local resonance interplay and enables ultra-low frequency bandgaps in the triangular lattice variant [179].

3.4. Thermal management

Thermal management in heterogeneous metamaterials exploits material, geometric, and mechanism level heterogeneity to actively programme coefficients of thermal expansion (CTE) and couple thermomechanical responses with stiffness, density, and Poisson's ratio. Figure 12a introduces windmill-shaped negative thermal expansion metamaterials in which bending-dominated and stretching-dominated deformation mechanisms are deliberately embedded within the same topological motif. By tuning the geometric parameters of the windmill arms and their load-transfer pathways, thermal mismatch is converted into controlled rotational and bending motions, enabling the thermal expansion coefficient to be continuously tuned from negative to positive values. Figure 12b demonstrates a 3D metallic lattice whose unit cell geometry and composition co-design yields simultaneous negative/near-zero CTE and NPR while preserving high stiffness and low density; beam and solid FE models bracket the thermomechanical response [181]. This heterogeneous design concept, detailed in Figure 12c, originates from a 2D re-entrant unit cell composed of two distinct materials. These modules are assembled into 3D 'star shaped' and 'cubic' cells and then periodically arrayed. The resulting lattice exhibits a unique combination of a negative Poisson's ratio and a tunable, near-zero coefficient of thermal expansion (CTE) [182]. Figure 12d uses multimaterial 3D printing to interweave high and low CTE phases in a lightweight lattice so that thermal mismatch drives a controlled 'buckling reversal,' producing macroscopic contraction on heating, an architected, phase interlocking route to NTE like behaviour [183-185]. Finally, Figure 12e combines two metals of contrasting CTEs within a star-shaped re-entrant lattice, achieving dual negativity (NPR and NTE) confirmed by beam element FE and deformation maps: lateral contraction under compression and global contraction on heating, demonstrating that multi-material, multimechanism heterogeneity is the key lever for architected thermal responses [186,187].

3.5. Smart responsiveness

Smart metamaterials achieve programmability and adaptability by exploiting heterogeneity in multiple phases, architectures, and active elements, in a tightly coupled structure–mechanism–property loop. In Figure 13a, a cage-shaped self-folding mechanical metamaterial is introduced, in which temperature-responsive building blocks with distinct folding thresholds enable sequential, multi-mode shape transformation from an initially planar configuration. By programming heterogeneous thermal activation and folding pathways, the structure achieves stable multistage deformation, tunable stiffness, and decoupled load-bearing responses, illustrating how material heterogeneity and self-folding kinematics can be combined for programmable mechanical functionality [188]. In Figure 13b, mechanical metamaterials composed of two 3D-printable polymers with mismatched temperature-dependent moduli realise shape memory–like behaviour without relying on intrinsic shape memory effects. The rational interconnection of these materials enables thermally triggered, mechanically governed multistability and large reversible deformations, eliminating conventional thermomechanical programming while achieving sequential shape reconfiguration and robust recovery through heterogeneous material coupling [190]. In Figure 13c, robust bonding of hydrogels to UV curable elastomers/rigids via multimaterial 3D printing yields graded stiffness hybrids suitable for bio integrated devices (e.g. artificial menisci) [191]. Figure 13d integrates a passive metallic patch array with an active non foster NIC to form a hybrid meta surface that breaks the passive narrow band limit and expands

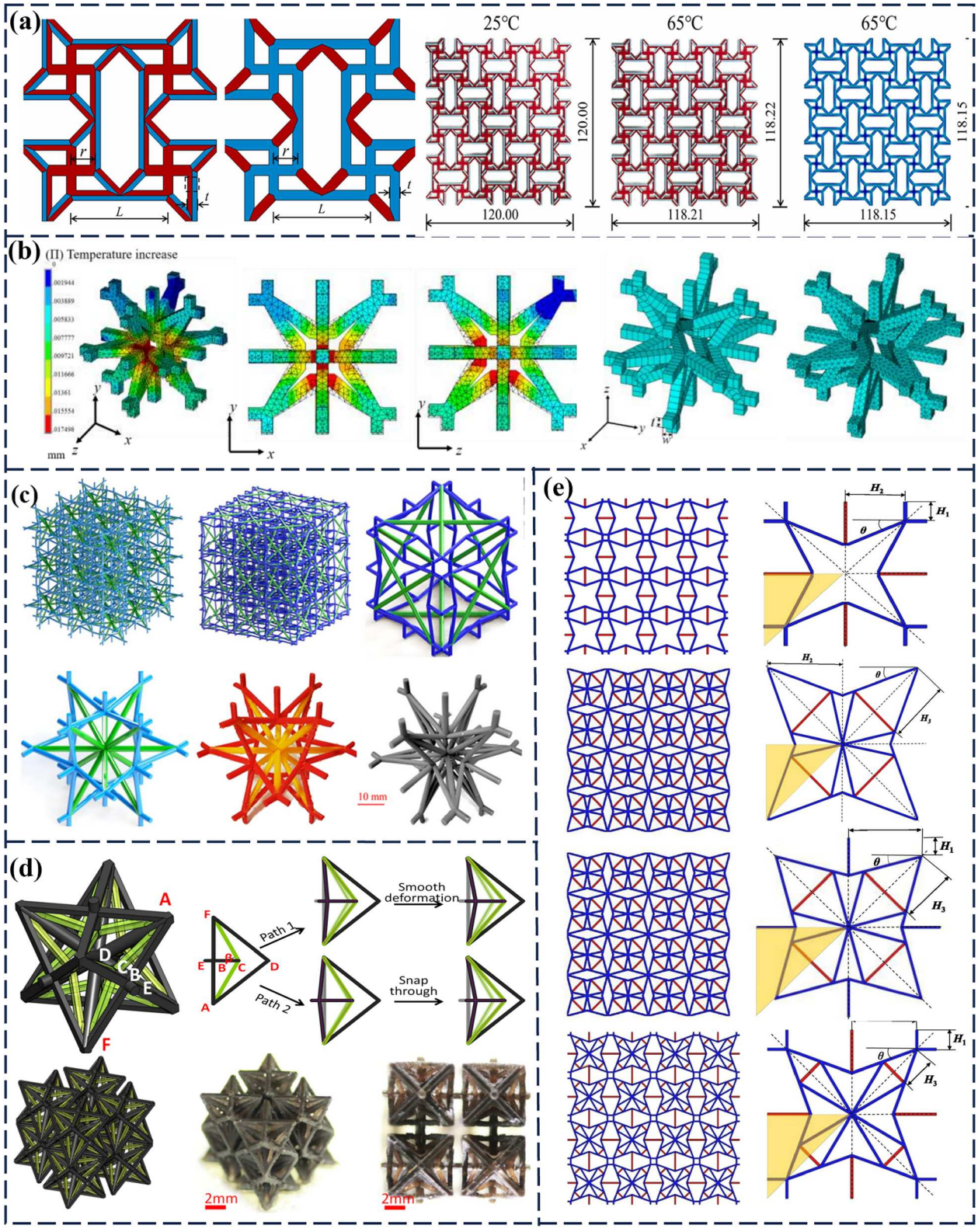


Figure 12. A tunable coefficient of thermal expansion: (a) Windmill-shaped negative thermal expansion metamaterial [180]; (b) Thermomechanical deformation of 3D metallic lattice [181]; (c) Geometry of bi material meta lattices [182]; (d) Multimaterial interwoven lattice, CAD and fabricated unit cells (3D/2D views) [183]; (e) Bi-metal re-entrant/star unit cell for dual negativity [186].

cloaking bandwidth by an order of magnitude [192]. Figure 13e leverages a two-phase prestressed stack (deformer + constrainer) that thermally self-assembles into ultralight, super-elastic lattices with load-adaptive stiffness and double the recoverable strain of prior systems [193]. Finally, Figure 13f uses origami inspired

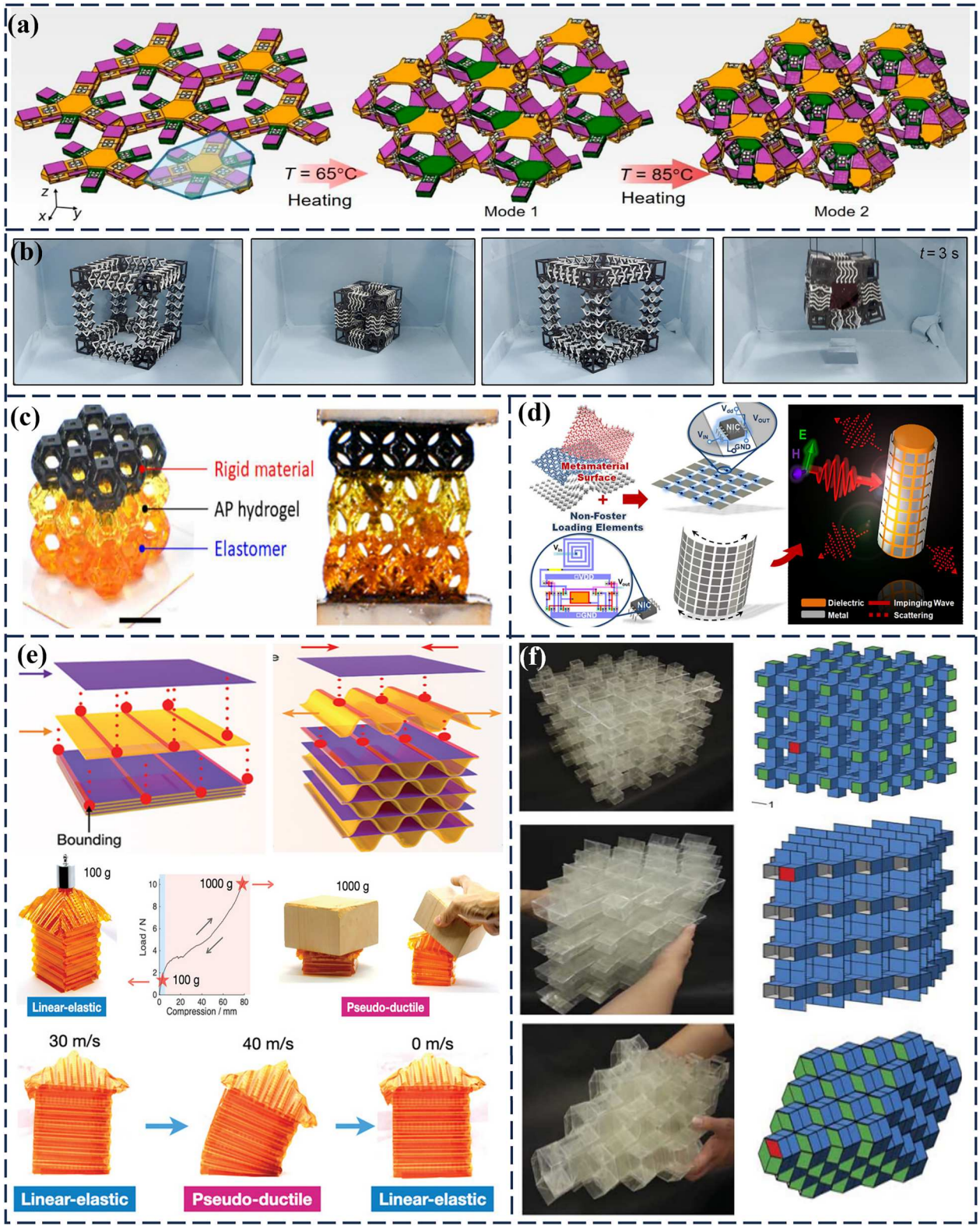


Figure 13. Smart heterogeneous metamaterials, (a) Thermally activated self-folding smart metamaterial [188]; (b) Multi-material smart metamaterial enabling programmable shape memory and multistability via modulus mismatch [189]; (c) Multimaterial 3D printed hybrids (Kelvin foam, graded stiffness cubes, reinforced hydrogels) [190]; (d) Hybrid non-Foster metasurface cloak with bandwidth expansion [191]; (e) Two phase prestressed stacks yielding ultralight super elastic lattices [192]; (f) Topology-optimised bistable units with programmed snap through sequences [193]; (h) Origami inspired reconfigurable acoustic waveguides [194].

rigid panels and elastic hinges to reconfigure internal channel connectivity/orientation, enabling actively switchable acoustic waveguides, a functional manifestation of geometric and material heterogeneity [194]. Collectively, Figure 13 demonstrates that biphasic and multimaterial heterogeneity, augmented by active elements and programmable geometry, is the central enabler of smart, reconfigurable responses [195].

3.6. Bistable characteristics

Heterogeneous bistable metamaterials deliberately embed internal non-uniformity in material phases, geometry, and inter-unit coupling to engineer an energy landscape with multiple metastable wells and controllable transition pathways. Figure 14(a) introduces the multistable building block, a geometrically programmed unit that undergoes snap through between discrete equilibria, establishing mechanism level heterogeneity (local curvature, ligament thickness, compliant hinges) that tunes barrier height, switching force, and rate sensitivity [196]. Figure 14(b) exploits material heterogeneity to emulate SMA-like superelasticity and shape memory through a bistable sinusoidal beam embedded in supports with strongly temperature-dependent stiffness. The mismatch in storage modulus evolution reshapes the local energy landscape, enabling load-induced snap-through between two stable configurations at low temperature and autonomous recovery to the primary state above a critical temperature, thereby translating phase-dependent material contrast into thermally programmed stability switching at the unit-cell level [197]. The macroscopic constitutive behaviour in Figure 14(c) arises from the cooperative activation of many bistable cells; progressive snap-through generates a broad force plateau and enhanced specific energy absorption at near-constant load, while programmable unit statistics (threshold dispersion, coupling strength) regulate plateau width, slope, and re-entrant segments critical for impact mitigation and vibration isolation [198]. Figure 14(d) validates the dynamic operating principle: under compression, elastic energy is stored and then released abruptly as snap through propagates; combined high speed experiments and simulations quantify inertia–damping competition, switching fronts, and cycle to cycle stability, key for fatigue life and rate dependent performance in strongly nonlinear regimes [199]. Figure 14(e) extends this concept by constructing a multistable metamaterial from bi-material sinusoidal beams whose stability topology is governed by temperature. A phase diagram maps the transition from bistability to monostability, allowing programmable snap-through under mechanical loading followed by temperature-driven recovery [200]. Collective activation of such units produces temperature-induced recoverability and reconfigurable deformation pathways, with tunable triggering thresholds enabling sequential and multi-stage stability control. Figure 14(f) introduces granular metamaterials composed of discrete bimaterial particles that undergo reversible bond formation and rupture under combined mechanical and thermal stimuli. Unlike lattice-based systems with fixed connectivity, dynamic inter-particle coupling enables bond reconfiguration, allowing the material to switch between assembled and disassembled states. This contact-mediated heterogeneity supports large, reversible changes in stiffness and deformation mode, and gives rise to collective behaviours such as directional transport, object capture, and adaptive shape reconfiguration [201]. Together, Figure 14 advance a unifying view in which heterogeneity, at the unit, network, and actuation levels, governs the structure, mechanism–property loop, enabling programmable plateaus, robust dissipation, and intelligent control that exceed the limits of homogeneous multistable systems.

4. Future properties

4.1. Key technical challenges

The rapid development of heterogeneous mechanical metamaterials has opened up new avenues for advanced material systems with exceptional mechanical and multifunctional properties [202 203]. However, several key technical challenges must be addressed to fully realise the potential of these materials in practical engineering applications.

First of all, in terms of the preparation technology of heterostructural metamaterials. The preparation technology of heterogeneous metamaterials is still the main technical difficulty restricting the practical use of metamaterials in engineering and equipment, and the current research work is mainly limited to academic research, focusing on new hybrid configurations, multifunctional integration and regulation, deformation mechanism and failure mechanism, but the impact of structural manufacturing defects on long-term service performance is significantly insufficient [204]. Because the size of the superstructure feature (about 0.1-10 mm) only contains a few to dozens of grains, the roughness and internal defects of the sample surface are still relatively few studies on the macroscopic service performance, mainly due to the lack of high-precision preparation and processing methods and service performance evaluation methods considering

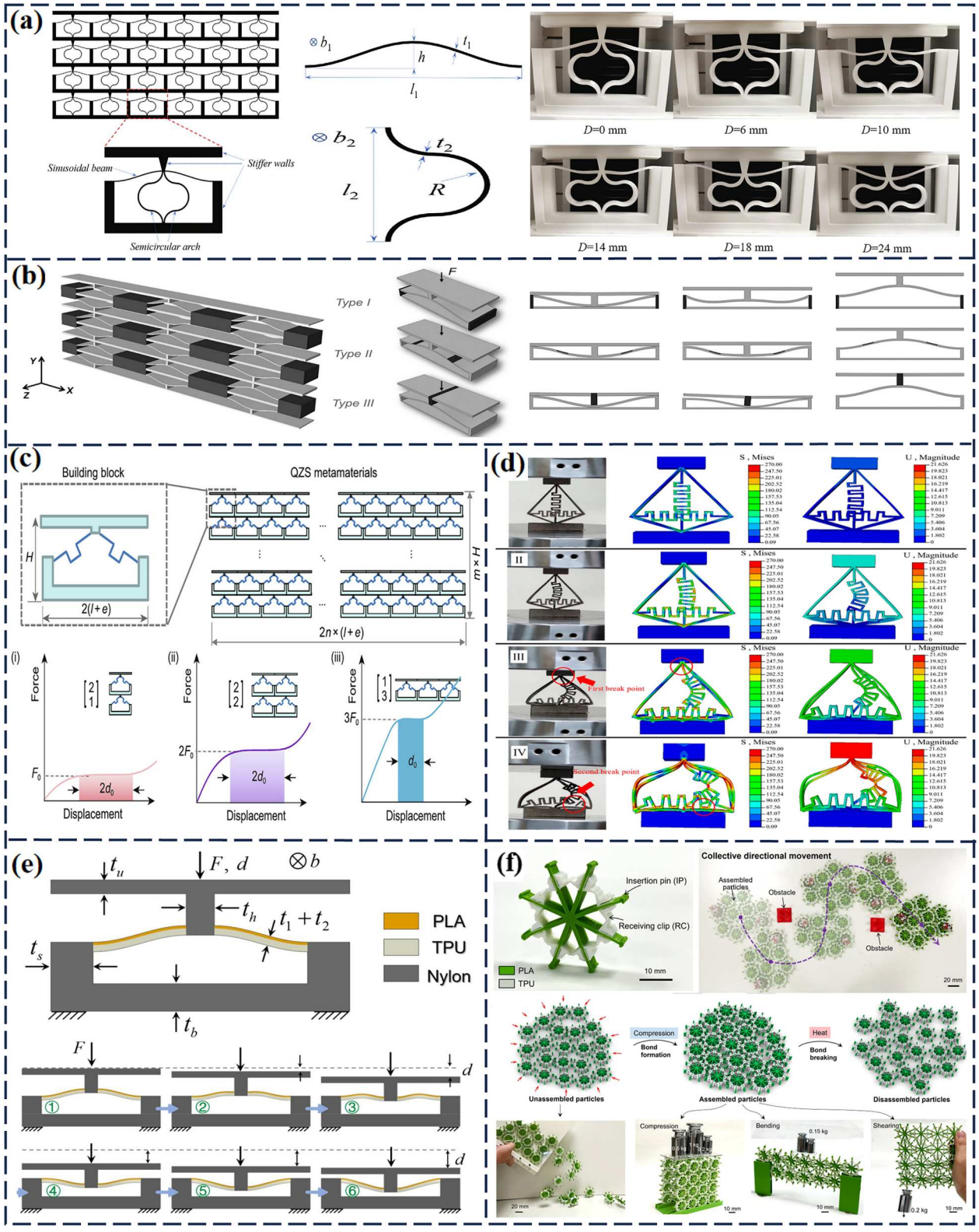


Figure 14. Bistable characteristics of heterogeneous metamaterials, (a) PR sign switching unit enabling programmable stiffness/NPR [196]; (b) Temperature-programmed bistable unit enabling SMA-like multistable responses [197]; (c) Programmable QZS metamaterials with tailored force–displacement plateaus [198]; (d) Laser printed Miura origami tube achieving QZS and strong longitudinal wave control [199]; (e) Phase-diagram-guided multistable metamaterial with programmable snap-through [200]; (f) Granular multistable metamaterial enabled by dynamic bond reconfiguration [201].

manufacturing defects [205], particularly for multi-material and multi-scale designs. In multi-material printing, reliable performance is often limited by material compatibility and interfacial quality, as mismatches in rheology, curing kinetics, or thermal response can induce weak bonding, interfacial porosity, and material cross-contamination. These defects degrade interfacial load transfer and frequently act as preferential sites for damage initiation. At the multi-scale level, achieving simultaneous control over fine architectural features and large structural dimensions remains challenging due to intrinsic trade-offs among printing resolution, build volume, and processing time. In addition, defect accumulation across length scales can compromise mechanical consistency, making it difficult to preserve designed heterogeneity when scaling from unit-cell demonstrations to system-level structures [206]. Therefore, the further development of heterogeneous metamaterials and their engineering applications strongly relies on high-precision material processing and preparation technologies at various scales.

Secondly, experimental characterisation faces intrinsic limitations, particularly for in-situ interrogation of internal deformation mechanisms. For heterogeneous metamaterials, synchronised full-field strain mapping across different phases remains challenging, as mismatches in optical contrast, stiffness, and deformation modes complicate the application of conventional surface-based techniques such as digital image correlation, especially near material interfaces. Moreover, non-destructive observation of internal deformation, contact evolution, and interfacial debonding within three-dimensional heterogeneous structures is still constrained by the spatial and temporal resolution of current in-situ X-ray or neutron imaging methods, as well as by limited material contrast between constituent phases. These constraints restrict direct experimental access to localised failure initiation and progressive damage evolution, underscoring the gap between theoretically predicted deformation mechanisms and experimentally observable responses in complex heterogeneous systems. Furthermore, the types of basic structure primitives and cell elements of heterogeneous metamaterials, and the various interfacial performance parameters of heterostructures make it difficult to construct standardised test and characterisation techniques. In addition, large-scale multiscale heterostructure mechanical metamaterials contain thousands of structural primitives, and how to carry out cross-scale experiments of different structural components and cell elements to characterise the transfer of parameters between different structural scales is still a technical bottleneck of strong nonlinearity, lack of equations, and few parameters. There is still a lack of effective means for service performance monitoring under the working environmental conditions of multifield coupling or extreme service (strong dynamic load, ultra-high temperature, etc.).

Finally, the development of heterostructure mechanical metamaterials engineering technology is relatively lagging behind, and the technological maturity is insufficient. Metamaterials have a higher degree of design freedom and performance space than conventional materials, and fabricating metamaterial artificial structures often requires more complex technologies. For example, the feature size and accuracy of mechanical metamaterials in additive manufacturing heterostructures are limited by the process, and the overall size of the prepared components is also limited, which makes it difficult to achieve large-scale, low-cost, and high-efficiency manufacturing, and restricting the industrialisation of mechanical metamaterials and the application of large quantities and wide areas.

4.2. Development trends

As research progresses, future development trends in heterogeneous metamaterials will focus on the integration of structural components for multifunctional purposes. It is of critical importance to carry out multifunctional integrated design of structural components for major industrial equipment, such as lightweight and high load bearing integration, lightweight and high temperature resistance integration, lightweight and robust radiation resistance integration, lightweight and harsh corrosion resistance integration, lightweight and long term fatigue performance integration, lightweight and strong oxidation resistance integration, lightweight and high impact resistance, vibration insulation, and low thermal expansion integration properties. Besides, it is mandatory to carry out structural safety and durability performance evaluation of advanced mechanostructures through digital twin technique and structural components service performance verification experiments.

As shown in Figure15, following key steps are included for tailing the multiscale structural features of mechanostructures: (a) Industrial Equipment Demands: Structural genome composition and unit cell design, screening and modification regulation optimisation of metamaterials, lightweight, high load bearing, high temperature resistance, corrosion resistance, fatigue resistance, oxidation resistance, impact resistance, vibration insulation, sound absorption and noise reduction, wave absorption and wave transmission functions customisation; (b) Structural genome library:

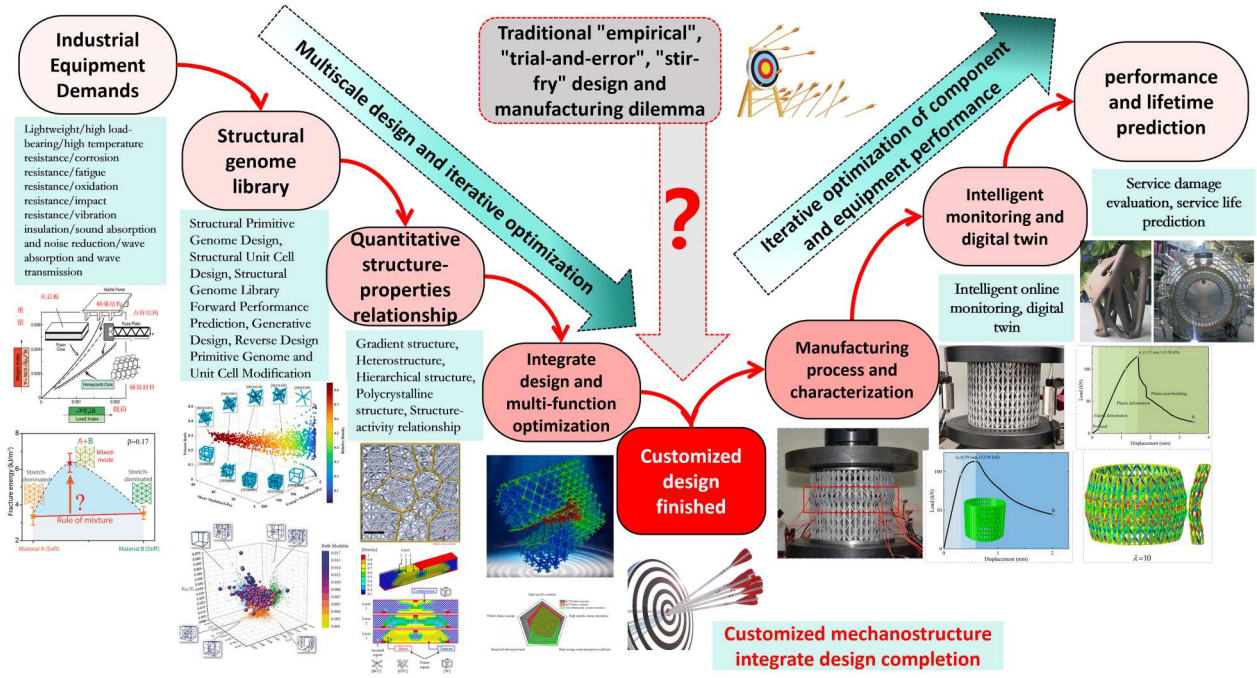


Figure 15. Future development trends in heterogeneous metamaterials: targeted customisation, efficient manufacturing, structural health monitoring, and service evaluation of cross scale composites and structures [207–229].

Structural Primitive Genome Design, Structural Unit Cell Design, Structural Genome Library Forward Performance Prediction, Generative Design, Reverse Design, Primitive Genome and Unit Cell Modification; (c) Quantitative structure properties relationship: gradient structure, heterostructure, hierarchical structure, polycrystalline structure, structure activity relationship; (d) Integrating design and multifunction optimisation; (e) Customised design is finished; (f) Manufacturing process and characterisation; (g) Intelligent monitoring and digital twin; (h) performance and lifetime prediction. The term Structural Genome is used to describe a parameterised representation of heterogeneous mechanical metamaterials that encodes the essential architectural information, including topology, geometry, material distribution, interface characteristics, and spatial variations, in a form suitable for systematic exploration and optimisation. The Primitive Genome refers to the minimal set of fundamental building units, such as basic strut or shell elements, node or joint types, and local interface motifs, from which more complex heterogeneous architectures are constructed. To improve the reliability of multi-scale modelling and the validation of data-driven design, standardised benchmark frameworks are required. Such benchmarks should consist of clearly defined unit-cell families, boundary conditions, loading protocols, and reference material systems, so that different numerical models and learning-based approaches can be evaluated on a common basis. In parallel, benchmark architectures should be accompanied by shared datasets that couple geometric descriptors and process parameters with experimentally validated mechanical responses, thereby providing reproducible physical ground truth for data-driven models. Cross-scale validation pipelines are further needed, in which reduced-order or homogenised models are calibrated and assessed against full-field simulations and targeted experiments performed on the same benchmark structures, ensuring consistency across length scales and modelling fidelities.

In conclusion, these development trends highlight a significant shift towards designing heterogeneous metamaterials that meet the increasing demands for multifunctional properties in complex industrial applications. The integration of advanced design strategies, data-driven optimisation methods, and cutting-edge manufacturing techniques will enable the creation of materials with unprecedented performance characteristics, tailored for specific operational conditions. The utilisation of digital twin technologies and intelligent monitoring will ensure that these advanced materials maintain their durability and performance in real-world applications.

5. Conclusion

Heterogeneous mechanical metamaterials represent a transformative leap in material science, offering

opportunities to overcome the inherent limitations of traditional homogeneous materials. These materials, which integrate diverse material phases, multiscale structural hierarchies, and multifunctional properties, have the potential to revolutionise a wide array of engineering applications, from impact protection and energy absorption to advanced thermal management and adaptive systems. This review has systematically explored the latest design strategies for heterogeneous mechanical metamaterials, including multiphase, hybrid architectures, interpenetrating phase composites, multimaterial integration, and multifeature fusion. Each design strategy has been critically assessed, emphasising the synergistic coupling between material heterogeneity, structural configuration, and enhanced performance. Through the exploration of these design methodologies, we have highlighted the complex interplay of mechanical behaviours, deformation mechanisms, and multifunctional capabilities that distinguish heterogeneous systems from conventional materials.

Despite the significant progress made in the development of heterogeneous metamaterials, several critical challenges remain. First, the fundamental understanding of structure-mechanism-property relationships, particularly in terms of interfacial regulation, failure evolution, and the functional coupling of heterogeneous materials, is still underdeveloped. Second, the scalability of manufacturing processes, coupled with the difficulty of achieving precision in the fabrication of multiphase materials, presents substantial hurdles in transitioning from academic research to practical, large-scale applications. Additionally, issues related to testing, characterisation, and the performance of these materials under real world conditions remain a barrier to their widespread implementation. Looking forward, the future of heterogeneous metamaterials will be shaped by the integration of multifunctional capabilities, data driven design, and advanced manufacturing technologies. The incorporation of digital twin technologies and real-time monitoring systems will enable more precise predictions of material behaviour and performance in complex service environments. Moreover, the continued refinement of multiscale modelling techniques and the development of high throughput characterisation methods will be essential to unlocking the full potential of these materials.

In conclusion, heterogeneous mechanical metamaterials stand poised to play a pivotal role in the next generation of engineering materials. By leveraging the inherent complexity and tunability of these systems, we can design materials with unprecedented performance characteristics, tailored to meet the diverse and evolving demands of modern industrial applications. As research progresses and these challenges are addressed, heterogeneous metamaterials will undoubtedly become a cornerstone of advanced engineering solutions, driving innovation in fields ranging from aerospace and defense to energy systems and beyond.

Author contributions

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Data availability statement

The data supporting the findings of this study are openly available in Figshare at https://doi.org/10.6084/m9.figshare.31029529.

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